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**Numerical Simulation of Ammonia Production: Comparison
of Conventional Route and Linde Concept.**

Dissertação de Mestrado

Dissertation presented to the Programa de Pós-Graduação em Engenharia Química, de Materiais e Processos Ambientais in partial fulfillment of the requirements for the degree of Mestre em Engenharia Química, de Materiais e Processos Ambientais.

Advisor: Prof. Amanda Lemette Teixeira Brandão.

Co-Advisor: Prof. Roberto Bentes de Carvalho.

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Abstract

Oliveira, Juan; Lemette, Amanda; Bentes, Roberto. **Numerical Simulation of Ammonia Production: Comparison of Conventional Route and Linde Concept**. Rio de Janeiro, 2024. 100p. Masters Dissertation—Departamento de Engenharia Química e de Materiais, Pontifícia Universidade Católica do Rio de Janeiro.

Ammonia is a globally traded chemical with an estimated market value of USD 79.63 billion in 2022, projected to reach USD 113.27 billion by 2030, driven primarily by its role in the fertilizer industry. Its high volumetric energy density and applications in refrigeration, power generation, and hydrogen storage make it strategically important. Due to its versatility, ammonia remains a cornerstone of the chemical industry and a focus of ongoing technological innovation. This dissertation presents a comparative simulation-based study of the Haber-Bosch (HB) process and the Linde Ammonia Concept (LAC), which incorporates Pressure Swing Adsorption (PSA) and an Air Separation Unit (ASU). Using Aspen HYSYS, each route was modeled across key units—reforming, WGS, CO₂ capture, purification, and synthesis—applying kinetic and thermodynamic models to reactor behavior, supported by sensitivity analysis and operating condition optimization. Several reactor configurations were analyzed, including adiabatic, autothermal, and multi-bed setups with nitrogen excess. The LAC process outperformed HB in productivity and energy efficiency, achieving 665.1 kmol/h of ammonia versus 578.0 kmol/h, and consuming 40.25 GJ/t NH₃ compared to 41.46 GJ/t NH₃ for HB. These gains stem from LAC's simplified flow, PSA purification, and optimized nitrogen strategies. Its design facilitates integration of advanced control schemes and modular operation, reaffirming LAC as a technically and energetically superior alternative for sustainable ammonia production.

Keywords:

Simulation and Optimization; Ammonia Synthesis; Methane Reforming; Aspen Hysys; Chemical Process.

Resumo

Oliveira, Juan; Lemette, Amanda; Bentes, Roberto. **Simulação Numérica da Produção de Amônia: Comparação da Rota Convencional e o Conceito Linde**. Rio de Janeiro, 2024. 100p. Dissertação de Mestrado – Departamento de Engenharia Química e de Materiais, Pontifícia Universidade Católica do Rio de Janeiro.

A amônia é um produto químico comercializado globalmente, com valor de mercado estimado em USD 79,63 bilhões em 2022 e projeção de atingir USD 113,27 bilhões até 2030, impulsionado principalmente por seu papel na indústria de fertilizantes. Sua alta densidade energética volumétrica e aplicações em refrigeração, geração de energia e armazenamento de hidrogênio a tornam estrategicamente relevante. Devido à versatilidade, a amônia segue como base da indústria química e foco de contínua inovação tecnológica. Esta dissertação apresenta um estudo comparativo, baseado em simulação, dos processos Haber-Bosch (HB) e Linde Ammonia Concept (LAC), que incorpora Adsorção por Oscilação de Pressão (PSA) e uma Unidade de Separação de Ar (ASU). Utilizando o Aspen HYSYS, cada rota foi modelada em unidades-chave — reforma, WGS, captura de CO₂, purificação e síntese — aplicando modelos cinéticos e termodinâmicos ao comportamento dos reatores, com suporte de análise de sensibilidade e otimização operacional. Foram analisadas várias configurações de reatores, incluindo sistemas adiabáticos, autotérmicos e multileito com excesso de nitrogênio. O processo LAC superou o HB em produtividade e eficiência energética, alcançando 665,1 kmol/h de amônia frente a 578,0 kmol/h, e consumindo 40,25 GJ/t NH₃ contra 41,46 GJ/t NH₃ no HB. Esses ganhos resultam do fluxo simplificado do LAC, da purificação por PSA e das estratégias otimizadas de nitrogênio. Seu projeto facilita a integração de esquemas avançados de controle e operação modular, reafirmando o LAC como alternativa técnica e energeticamente superior para uma produção sustentável de amônia.

Palavras-chave:

Simulação e otimização; Síntese de Amônia; Reforma do Metano; Aspen Hysys; Processo Químico.

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Abbreviations List

APP – Ammonium Polyphosphate

ASU – Air Separation Unit

ATR – Autothermal Reforming

CAPEX – Capital Expenditure

CSM – Activated Carbon Sieve

DEPG – Dimethyl Ethers of Polyethylene Glycol (Selexol)

HB – Haber-Bosch

HTS – High Temperature Shift

HT-WGSR – High Temperature Water Gas Shift Reactor

LAC – Linde Ammonia Concept

LPG – Liquified Petroleum Gas

LTS – Low Temperature Shift

LT-WGSR – Low Temperature Water Gas Shift Reactor

MA-ATR - Membrane-Assisted Autothermal Reforming

MAP – Mono-Ammonium Phosphate

MEA – Monoethanolamine

MTS – Medium Temperature Shift

OPEX – Operating Expenditure

PFR – Plug Flow Reactor

POM – Partial Oxidation of Methane

PR – Peng-Robinson

PSA – Pressure Swing Adsorption

SEG - Sorption-Enhanced Gasification

SMR – Steam Methane Reforming

WGS – Water Gas Shift

Symbols List

$C_{p,c}$ – Specific heat capacity of catalyst

E_{a1} – Activation Energy of direct reaction per kmol of N_2

E_{a2} – Activation Energy of reverse reaction per kmol of N_2

K_{eq} – Equilibrium Constant

Q_t – Total Energy Spent

T_{in} – Temperature of inlet stream

T_{out} – Temperature of outlet stream

T_r – Reactor Temperature

V_r – Reactor Volume

X – Reactor Conversion

X_g – Reaction Global Conversion

f – Catalyst activity factor

$k_{0,1}$ – Pre-exponential Arrhenius factor of direct reaction

$k_{0,2}$ – Pre-exponential Arrhenius factor of reverse reaction

ε – Void fraction

p_c – Packing density of catalyst

Try to move the world – the first step will be to move yourself.
Plato

1.

Introduction

Ammonia is a globally traded and highly valuable product, with its primary application being in the fertilizer sector (Guo; Wang; Li, 2023; Moura, I. P. *et al.*, 2021; Rodrigues *et al.*, 2022). It serves as a crucial chemical feedstock and finds application in refrigeration systems and power generation. Notably, its high volumetric energy density enables the transportation of nearly 50% more energy compared to Liquefied Petroleum Gas (LPG) through the same pipelines. Furthermore, ammonia can be utilized as a means for hydrogen storage (Fúnez Guerra *et al.*, 2020). The global ammonia market was valued at \$79.63 billion in 2022, with an expected growth at a CAGR of 4.07% between 2023 and 2030, reaching an estimated value of \$113.27 billion (PRODUCTION, CONSUMPTION AND TRADE OF AMMONIA IN SELECTED COUNTRIES AND REGIONS, 2020, 2022). Its versatility and energetic value make it an important compound. Although ammonia can be directly used as an inorganic fertilizer, it plays a crucial role in the production of nitrogen-based fertilizers such as urea (Moura, I. P. *et al.*, 2021), mono-ammonium phosphate (MAP), ammonium polyphosphate (APP) (Guo; Wang; Li, 2023), nitrate, and sulfate (Rodrigues *et al.*, 2022). In 2020, the largest ammonia-producing countries were China, Russia, the United States, and India, with respective outputs of 54 Mt, 20 Mt, 17 Mt, and 15 Mt (Shantanu S., 2024).

Ammonia is a globally traded and highly valuable product utilized across various industrial sectors. It serves as a crucial chemical feedstock, finds application in refrigeration systems, power generation. Notably, its high volumetric energy density enables the transportation of nearly 50% more energy compared to Liquefied Petroleum Gas (LPG) through the same pipelines. Furthermore, ammonia can be utilized as a means for hydrogen storage (Fúnez Guerra *et al.*, 2020). Its versatility and energetic value make it an important compound, with its primary application being in the fertilizer sector. Although ammonia can be directly used as an inorganic fertilizer, it plays a crucial role in the production of nitrogen-based fertilizers such as urea (Moura, I. P. *et al.*, 2021), mono-ammonium phosphate (MAP), ammonium polyphosphate (APP) (Guo; Wang; Li, 2023), nitrate and sulfate (Rodrigues *et al.*, 2022).

The most common method for synthetic ammonia production involves the heterogeneous reaction of atmospheric nitrogen and hydrogen using a surface-based iron catalyst at 200 atm and 500 °C - 600 °C (Fúnez Guerra *et al.*, 2020). While nitrogen gas can be obtained from the air, the main challenge lies in hydrogen production. Various pathways exist for hydrogen production, including steam reforming of hydrocarbons, coal gasification, enzymatic decomposition of sugar, conversion of glucose and alcohol, and electrolysis of water (Abbas; Dupont; Mahmud, 2017). The Congressional Research Service reported an overview of hydrogen production methods, highlighting that globally, 76% of hydrogen is produced from natural gas, 22% from coal, and 2% via water electrolysis. In the United States, the distribution is 95% from natural gas, 4% from coal, and 1% via water electrolysis. Currently, Steam Methane Reforming (SMR) remains the primary method employed due to its cost-effectiveness and the abundant availability of natural gas (Lexie, 2024). Additionally, methane comprises most of the Liquefied Petroleum Gas (LPG) composition along with molecules of sulfuric acid, nitrogen, and heavier alkanes (Ramdin *et al.*, 2016; Ramdin; De Loos; Vlugt, 2012), and it can also be found in biogas and shale gas and be obtained by biomass pyrolysis (Asencios *et al.*, 2016; Couto *et al.*, 2013).

In this work, hydrogen, obtained through methane reforming, plays a critical role as a raw material in fertilizer and ammonia production. Beyond this application, hydrogen serves as an essential co-reactant in numerous refinery processes (Abbas; Dupont; Mahmud, 2017). It is also widely recognized as the fuel of the future due to its ability to generate electricity without emitting harmful pollutants (Abbas; Dupont; Mahmud, 2017; Chiron; Patience; Riffart, 2011). When combusted, hydrogen produces only water vapor as a byproduct, releasing no greenhouse gases or pollutants such as SO_x, soot, or particulate matter into the environment. However, the reforming process used to produce hydrogen generates by-products such as carbon monoxide, carbon dioxide, and water vapor. A mixture of these gases is commercially known as SynGas (Rabe; Truong; Vogel, 2005), which has multiple industrial applications. SynGas can be used as a fuel in certain types of engines, as a reducing gas in the steel industry (Cavaliere; Perrone; Marsano, 2023), or in Fischer-Tropsch synthesis to produce alkanes, paraffins, and oxygenates such as aldehydes and ketones (Asencios *et al.*, 2016; Luis *et al.*, 2011).

In this work, the production of CO is important to enhance the production of hydrogen at the Water Gas Shift (WGS) stage. In this process, the CO is also converted into CO₂, thereby, the gas in the carbon dioxide form can be better absorbed by the appropriate solvents used in the absorption column. The by-product CO₂ is captured and stored as it can be applied for urea production later. This approach aligns with the blue hydrogen route, where carbon capture and storage technologies are employed to reduce greenhouse gas emissions associated with hydrogen production from fossil fuels (Massarweh *et al.*, 2023).

In ammonia plants, the objective is to generate syngas with a higher H₂/CO ratio and produce a greater quantity of the primary raw material. The steam methane reforming (SMR) reaction is the primary method for methane conversion, as it yields a H₂/CO molar ratio of 3:1, which is higher than that of other potential methane reforming reactions, thereby enhancing the rate of hydrogen production (Rabe; Truong; Vogel, 2005).

The Haber-Bosch process, developed by Fritz Haber and industrially implemented by Carl Bosch in 1913, remains the predominant method for ammonia production. Initially, methane undergoes steam reforming (SMR), producing hydrogen-rich synthesis gas. Next, the second reformer introduces air to facilitate combustion and partial oxidation of methane, generating additional hydrogen and the nitrogen necessary for ammonia synthesis. Subsequently, the synthesis gas passes through two Water Gas Shift Reactors—first at high temperature (HT-WGSR) and then at lower temperature (LT-WGSR)—to convert carbon monoxide into additional hydrogen and carbon dioxide. Optionally, a CO₂ capture section can be integrated, typically employing chemical absorption to remove carbon dioxide and avoid catalyst contamination. Afterward, a methanation stage purifies the synthesis gas further, eliminating residual carbon monoxide and carbon dioxide impurities through reaction with hydrogen, ensuring catalyst protection. Lastly, purified gas enters the ammonia reactor, where nitrogen and hydrogen react catalytically under high pressure and temperature, using an iron-based catalyst, ultimately forming ammonia (THE HABER-BOSCH HERITAGE, 1997).

Over the past century, new strategies have come to adapt the Haber-Bosch process, focusing on reducing energy consumption and enhancing productivity. Some examples of a

few changes are the addition of Pressure Swing Adsorption (PSA), Air Separation Unit (ASU), reformers with steam and air simultaneously (Pattabathula; Richardson, 2016).

One of the alternative routes is named “The Linde Ammonia Concept” (LAC). The LAC process shares many similarities with the Haber-Bosch process in several steps; however, the key distinction among them is the nitrogen introduction. Unlike the Haber-Bosch process that employs a second reformer, methane is solely reformed using steam, and nitrogen is introduced only in the ammonia reactor, located at the final stage of the plant, in the LAC process. Nitrogen is obtained through an air separation unit (ASU), which is responsible for separating nitrogen gas and oxygen from atmospheric air. Additionally, in the LAC process, methanation step is replaced by a pressure swing adsorption (PSA) process, a common method used for purifying hydrogen gas (Pattabathula; Richardson, 2016).

In this work, the aim is to examine both industrial pathways for ammonia production: the conventional route Haber-Bosch (HB) and Linde Ammonia Concept (LAC). Therefore, it is crucial to outline every step of each process, highlighting both similarities and differences in their operation. A comprehensive block flow diagram will be presented to simply illustrate the strategies and associated process operations (Figures 1.1 and 1.2). Subsequent subsections will provide a detailed explanation of each process, encompassing operation conditions, employed catalysts, reactions and reactors involved, as well as alternative approaches that have been explored.

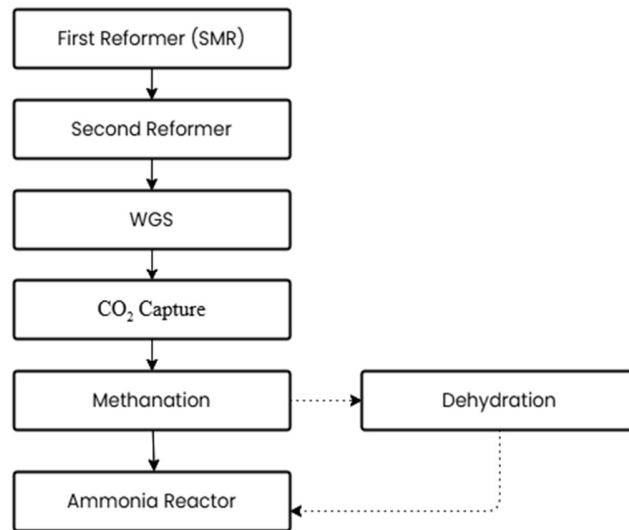


Figure 1.1: Scheme of the conventional Haber-Bosch (HB) ammonia process.

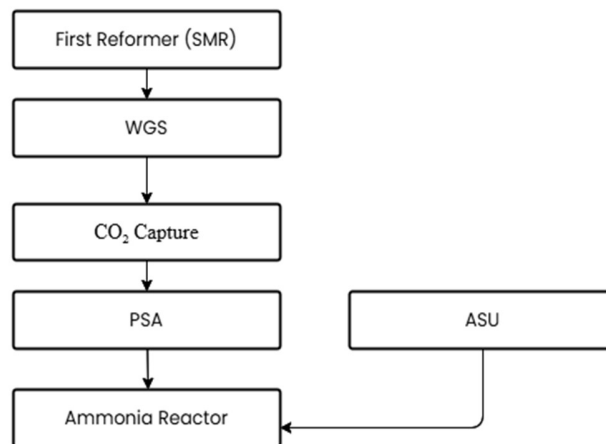


Figure 1.2: Scheme of Linde Ammonia Concept process (LAC).

Meanwhile, the Haber-Bosch process remains the predominant approach for ammonia production, accounting for 90% of global production (Deng *et al.*, 2022; Fúnez Guerra *et al.*, 2020). However, this process is associated with significant energy consumption, responsible for 2% of global energy usage, as well as contributing to 1% of

global greenhouse gas emissions in 2004 (Deng *et al.*, 2022). Given the current global context, there is growing concern and commitment towards addressing environmental issues, including the reduction of energy consumption and pollution (Quindimil *et al.*, 2022; Rau; Baird, 2018). Consequently, various alternatives are being explored to mitigate these challenges, such as modifying energy sources, incorporating a CO₂ sequestration sector to reduce emissions, and repurposing CO₂ co-product for future product development (Fúnez Guerra *et al.*, 2020).

This work aims to develop a detailed simulation-based analysis of HB and LAC routes. The specific objectives are: simulate the main reactions of each stage of the process, some based on kinetic aspects and others on thermodynamic equilibrium parameters; build sensitivity surfaces to study each stage of the process; optimize operating conditions; analyze the performance of CO₂ absorption at different temperatures and pressures and observe the behavior of energy consumption; test different ammonia reactor configurations, such as adiabatic, autothermal, and looping processes; test the insertion of excess nitrogen enabled by the ASU used in the LAC plant; compare the energy recovery capacity of each plant. Based on the results obtained from these analyses, the main objectives are: compare the two routes in terms of production; compare the two routes in terms of energy efficiency; choose the best route.

2.

Literature Review

The simulation-based studies reviewed in this chapter demonstrate a range of technical innovations and methodological approaches to optimize ammonia production through reforming, purification, and synthesis stages. Many of these studies directly coincide with the present dissertation, which investigates process performance through rigorous simulation in Aspen HYSYS and Aspen Plus.

One of the first studies to highlight the importance of production scale on ammonia plant performance is that of Pattabathula and Richardson (2016), which provides a comprehensive overview of technological evolution and energy benchmarks in conventional H-B plants. The correlation between production scale and energy efficiency has been well documented. (Pattabathula; Richardson, 2016), as well as Sagel et al. (2022), have demonstrated through simulation and industrial data analysis that smaller-scale ammonia production is associated with higher specific energy consumption, primarily due to lower thermal integration and the absence of scale-related efficiencies. These studies emphasize the challenges of decentralized ammonia synthesis in maintaining energy performance levels comparable to centralized, large-scale facilities. Supporting this, the study "Green ammonia enables sustainable energy and fertilizer production: A case study on the island of Curaçao" by Sagel et al. (2022) shows that small-scale plants (e.g., 1 ton NH_3 /day) may consume up to 50% more energy per tonne than plants operating at 50 ton/day, primarily due to the lower efficiency of electrolyzers and the limited potential for effective heat recovery in compact systems. Modern large-scale H-B plants typically achieve energy consumptions between 27 and 36 GJ per tonne of ammonia, approaching the theoretical minimum of 20.1 GJ/t (Rouwenhorst, Kevin *et al.*, 2020). However, such low specific energy values are only feasible at high production capacities, often above 1,500–3,300 t/day. This observation aligns with the findings of this dissertation, which confirm that smaller-scale ammonia production results in higher specific energy consumption due to reduced heat integration and lack of scale economies (Pattabathula; Richardson, 2016; Sagel; Rouwenhorst; Faria, 2022)

The reforming stages remain the most energy-intensive parts of the process. Studies such as those and Rouwenhorst et al. (2020) indicate that, after the secondary reformer, residual CH_4 concentrations typically fall below 0.5 vol%, a value consistent with the simulation outcomes presented in this dissertation. Similarly, the low-temperature water-gas shift (WGS) stage achieves residual CO concentrations below 0.3 vol %, a benchmark also validated in the present study. From the ammonia production standpoint, the authors indicate that the Haber-Bosch process, conducted in high-pressure reactors operating typically between 100 and 250 bar, achieves an outlet ammonia concentration of approximately 25% by volume. This finding supports the conversion levels simulated in the present work, which implements a multi-bed reactor system with interstage cooling to replicate industrial loop configurations

An important thermodynamic study on the methanation reactions of CO and CO_2 is presented by Gao et al. (2012), which offers detailed insights into how temperature and pressure affect conversion and selectivity. Through simulation, it was observed that CO conversion begins to decline above 550 °C. They also found that increasing pressure up to 30 atm significantly improves CO conversion and methane selectivity, although benefits plateau beyond this point.

Alternative ammonia production strategies have also been extensively studied through simulation. One notable contribution is the study by Moura et al. (2021) , which explores an innovative route for ammonia and urea production based on methane bi-reforming (BRM) integrated with carbon dioxide capture from thermal power plant flue gas. This route aims to achieve significant CO_2 abatement while maintaining technical viability. Using Aspen Plus simulations, the authors compared this novel pathway with the conventional steam reforming route. Results demonstrated that the BRM-based route could abate up to 745 t/day of CO_2 , while the conventional route emits approximately 1.156 t/day. Although the hydrogen yield per unit of natural gas was lower in the BRM route (0.22 t H_2 /t NG vs. 0.41 in the conventional route), the BRM configuration required fewer processing units and achieved lower electricity consumption per product mass (0.27 MWh/t vs. 0.47 MWh/t), indicating improved energy utilization. Despite both configurations being economically unfeasible under base-case conditions, the BRM-based route showed better

economic indicators due to the generation of a CO-rich byproduct stream, which can be monetized.

Another alternative strategy explored in the literature is membrane-assisted autothermal reforming (MA-ATR), investigated by Cloete et al. (2021) as a cost-effective and energy-efficient route for CO₂-free ammonia synthesis. This process configuration integrates hydrogen production, heat supply, and CO₂ separation into a single reactor unit, reducing equipment count and simplifying flowsheet complexity. The simulation results showed a specific energy consumption of 25.8 GJ/t NH₃—lower than that of conventional H-B processes—and CO₂ emissions of just 0.18 t CO₂/t NH₃. Additionally, the simulation of the ammonia synthesis reactor in this study reported a hydrogen-to-ammonia conversion of 20.6% per pass, under operating conditions of 450 °C and 199 bar.

Simulation studies involving biomass gasification and sorption-enhanced reforming provide further evidence of the trade-off between energy performance and carbon footprint. In particular, the work of Sun et al. (2024) reports energy consumptions of 36.4 – 40.2 GJ/t NH₃ in small-scale green ammonia plants using Sorption-Enhanced Gasification (SEG). Notably, configurations with air separation units (ASUs) and CO₂ capture can achieve net-negative carbon emissions, though at the cost of added complexity. It is also important to note that, according to CO₂ capture studies using physical solvents such as Selexol, more than 70% of the energy demand in the CO₂ capture section is attributed to solvent regeneration (Belchior *et al.*, 2020)

Finally, the integration of process simulation with machine learning has emerged as a powerful tool for energy optimization in ammonia plants. According to El-Maghraby and Mohamed (2024), the author carried out simulations in Aspen HYSYS of four industrial ammonia plants, modeling separately the main energy consumers (reformer, boiler, synthesis gas compressor, and air compressor). Based on the simulated data, regression models were developed in Python to accurately predict specific energy consumption. This approach enabled the identification of key operational variables, resulting in an optimized operational strategy capable of reducing CO₂ emissions by 4.7 t/h and generating annual cost savings of

over USD 3.9 million. This demonstrates the potential of data-driven approaches to improve traditional processes, aligning with the optimization objectives proposed in this dissertation.

In summary, the reviewed literature supports several key aspects of this work: the relationship between plant scale and energy performance, benchmark values for residual species after reforming and WGS, and ammonia synthesis conversion efficiencies. Moreover, the simulation methodologies and alternative process configurations discussed in recent studies reinforce the relevance and applicability of the modeling strategies adopted in this dissertation.

3. Methodology

3.1

Preliminary Considerations

According to data provided by Yara, a large-scale industrial plant can produce approximately 800,000 tons of ammonia annually (YARA PILBARA FERTILIZERS, 2018). By applying a mass balance derived from Equations 2-3 and 2-9, an estimated maximum methane inlet flow rate of about 2,679.74 kmol/h is required to achieve this production capacity. However, considering feasibility constraints and aiming for practical operational conditions, a scaled-down approach was adopted for this study. Therefore, the methane feed rate was adjusted to approximately 2,679.74 kmol/h to a more manageable and realistic level. Consequently, an inlet methane flow rate of 2,679.74 kmol/h was scaled down by a factor of ten, resulting in a feasible operational flow rate of approximately 267.97 kmol/h.

The Peng–Robinson (PR) equation of state was used as the physical property analysis method. The equation of state is suitable for the calculation of gas–liquid thermodynamic properties and gas–liquid equilibrium of a variety of non-polar or weakly polar substances. Therefore, it is also suitable for the ammonia synthesis process (Deng *et al.*, 2022).

Two fluid packages were used in both ammonia plants: Peng-Robinson and Acid gas physical solvents. The first one is used to simulate all reactions, and the second specifically includes physical absorption mechanism and its solvents, such as DEPG (dimethyl ethers of polyethylene glycol). All components of Peng-Robinson besides ammonia are included in the CO₂ capture section (Table 3.1).

Table 3.1: Fluid Packages and components.

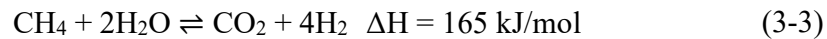
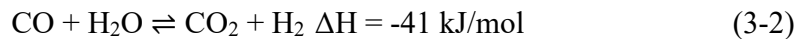
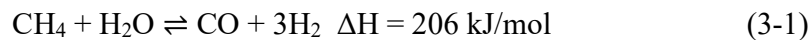
Peng-Robinson	Acid gas physical solvents
Methane	Methane
Water (steam)	Water (steam)
Carbon monoxide	Carbon monoxide
Hydrogen	Hydrogen
Carbon Dioxide	Carbon Dioxide

Nitrogen	Nitrogen
Argon	Argon
Oxygen	Oxygen
Ammonia	Hydrogen Sulfide
-	DEPG

3.2

HB Steps, Reactions and Modeling

The initial step of the Haber-Bosch process involves Steam Methane Reforming, which was simulated in an equilibrium reactor using thermodynamic parameters of two main independent reactions (Equations 3-1 and 3-2). The steam methane reformer is the reactor where methane and steam react to produce hydrogen, carbon monoxide and carbon dioxide. The SMR process combines the SMR reaction (Equation 3-1) and the Water Gas Shift reaction (Equation 3-2). Together, they result in the SMR Global Reaction (Equation 3-3).



At industrial conditions, the SMR process is typically operated within a temperature range of 800 - 1000 °C and at high pressures between 20 - 35 atm. It is important to note the endothermic nature of this reaction; therefore, achieving greater conversion requires a significant demand for energy to supply the reactor. Conventional reactors are typically characterized as furnace fields with tubes loaded with the appropriate metal catalyst (Abbas; Dupont; Mahmud, 2017).

Many researchers discovered different catalyst metals and catalyst supports to the SMR reaction. There are many active metals used to promote the SMR reaction, but nickel (Ni) has been widely used due to its high reactivity and low attrition. Ni-based catalysts are also advantageous for their low cost, high catalytic activity and non-pyrophoric nature, withstanding very high temperatures (900 - 1000 °C) (Abbas; Dupont; Mahmud, 2017;

Ashok; Wai; Kawi, 2018). Some other catalyst options could be cheaper, such as Manganese (Mn), Iron (Fe) and Copper (Cu), but compensating with a low percentage composition Nickel (Ni) is advisable. The optimal cost-benefit combination is nickel supported by alumina (α -Al₂O₃), which offers excellent reactivity, prevents agglomeration, and is the most widely used catalyst in industrial applications (Abbas; Dupont; Mahmud, 2017; Rydén; Lyngfelt; Mattisson, 2008). In literature, different values of steam/methane ratio are used to enter the reactor. For instance, Xu and Froment set the molar flow ratio to 3.4 over a Ni/MgAl₂O₄ catalyst (Xu; Froment, 1989). Meanwhile, this work uses a molar ratio value of 3.34 based on Devos and Froment study using Ni/a-Al₂O₃ catalyst (De Deken; Devos; Froment, 1982; Moura, I. P. *et al.*, 2021).

The conversion, component compositions, and heat transfer were calculated based on the equilibrium constant values obtained from the literature (Figures 3.1 and 3.2).

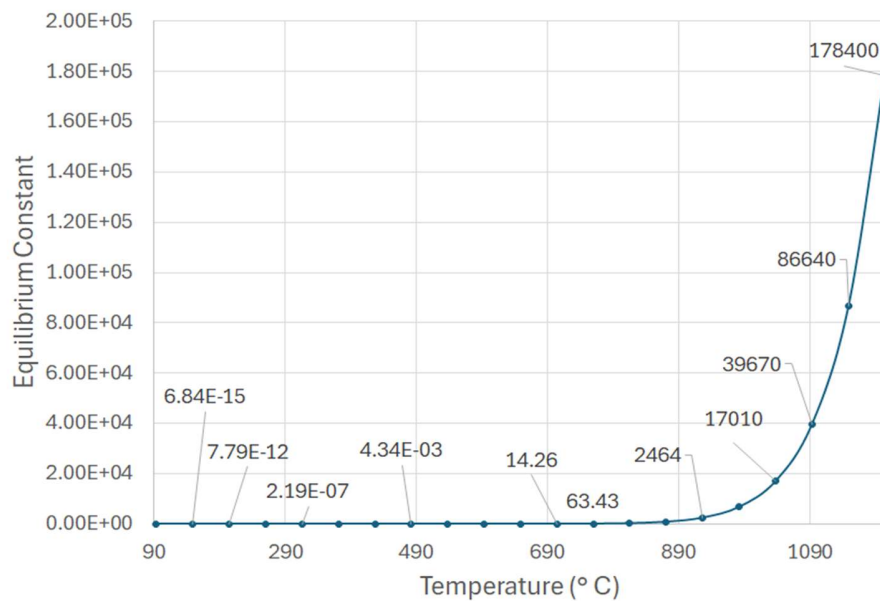


Figure 3.1: Relation between SMR equilibrium constant and temperature.

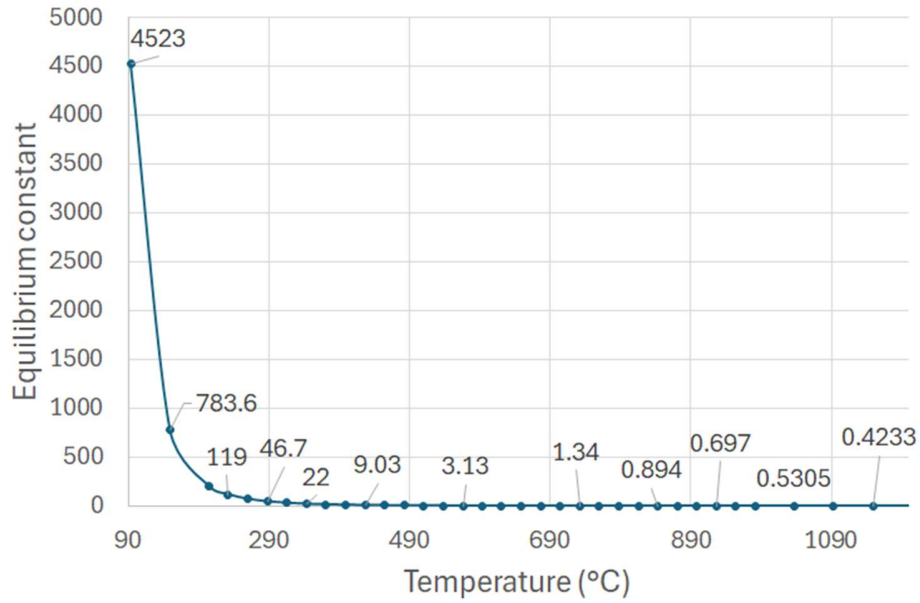
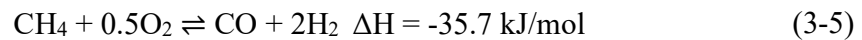
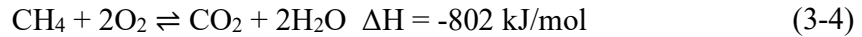


Figure 3. 2: Relation between WGS equilibrium constant and temperature.

The output from the first reformer is sent to the second reformer. Here, methane and oxygen undergo a reaction to produce more hydrogen and completely convert any remaining methane. Various strategies can be employed at this stage. The conventional approach is the autothermal reforming (ATR), which involves a combination of exothermic partial oxidation reaction (Equation 3-5), weakly exothermic water gas shift reaction (Equation. 3-2) and endothermic steam reforming reaction (Equation 3-1). This process can occur spontaneously, coupled with self-heating (Yan; Zhang; Zhang, 2013), typically operating at pressures ranging from 30 to 50 bar and temperatures between 950 and 1050 °C (Lamb *et al.*, 2020).



Another strategy involves the partial oxidation of methane (POM) (Equation 3-5), focusing on introducing pure O₂ below 700 °C (Rabe; Truong; Vogel, 2005) or operating at temperatures above 700 °C when using a catalyst due to methane and oxygen inertness (Prettre; Eichner; Perrin, 1946; Ren *et al.*, 2022). One great advantage of using pure oxygen

in this method is that O₂ is able to increase catalyst's lifetime reducing coke deposition (Asencios *et al.*, 2016).

In this study, the Jones-Lindstedt methane oxy-fuel combustion model was used to simulate the second reformer (Frassoldati *et al.*, 2009; Jain *et al.*, 2021; Jones; Lindstedt, 1988). A simplified version of this model, incorporating four reactions, is implemented to enhance computational efficiency and facilitate the conversion of simulations in Aspen Hysys. The model accounts for the kinetic rates of SMR (Equation 3-1) and POM (Equation 3-5) as direct reactions, while considering both direct and reverse reactions for WGS (Equation 3-2) and hydrogen combustion (Equation 3-6). The hydrogen combustion reaction is deemed undesirable, as it consumes hydrogen when proceeding in the forward direction. These four reactions represent independent equations that, through their linear combinations, facilitate the derivation of the additional reactions occurring in the second reformer. As an example, the methane complete oxidation reaction (Equation 3-4) is solved by a combination of Equations 3-2, 3-5 and 3-6.



The Table 3.2 presents the rate equations of the simplified Jones-Lindstead mechanism, considering POM and SMR as direct reactions and r_i and r'_i , respectively, the direct rate and the inverse rate of equation i ($i = 3-5, 3-1, 3-2, 3-6$) (Acampora; Marra, 2017; Frassoldati *et al.*, 2009), where T corresponds to temperature, R is the universal gas constant, and the equations are expressed in terms of the molar concentration of the components.

Table 3.2: Rate equations of simplified Jones-Lindstead mechanism.

Reaction	Rate equation	Units	
SMR (3-1)	$r_{3-1} = 3.84 \cdot 10^{12} e^{-\frac{30000}{RT}} [\text{CH}_4][\text{H}_2\text{O}]$	cm, s, cal, mol	(3-7)
WGS (3-2)	$r_{3-2} = 2.01 \cdot 10^{12} e^{-\frac{20000}{RT}} [\text{CO}][\text{H}_2\text{O}]$	cm, s, cal, mol	(3-8)

	$r'_{3-2} = 4.16057 \cdot 10^{17} e^{-\frac{30511.1798}{RT}} \cdot T^{-1.06114} [CO_2][H_2]$	cm, s, cal, mol	(3-9)
POM (3-5)	$r_{3-5} = 4.4 \cdot 10^{11} e^{-\frac{30000}{RT}} [CH_4]^{0.5} [O_2]^{1.25}$	l, s, cal, mol	(3-10)
Hydrogen combustion (3-6)	$r_{3-6} = 2.8492 \cdot 10^{19} e^{-\frac{40000}{RT}} \cdot T^{-1} [H_2]^{0.3} [O_2]^{1.55}$	cm, s, cal, mol	(3-11)
	$r'_{3-6} = 1.38099 \cdot 10^{19} e^{-\frac{97734.1716}{RT}} \cdot T^{-0.861783} [H_2]^{-0.7} [O_2]^{1.05} [H_2O]$	cm, s, cal, mol	(3-12)

Some researchers have reported the occurrence of autothermal reforming in small tubes similar to those used in the initial reforming process. The Jones-Lindstedt model can be employed to simulate this phenomenon (Jain *et al.*, 2021).

The third step of the Haber-Bosch process involves the Water Gas Shift (WGS) reactor, which plays a crucial role in producing hydrogen and carbon dioxide. The consumption of CO is essential to preserve the life of the Ni catalyst and prevent catalyst poisoning (Gardner; Bartholomew, 1981; Schmider; Maier; Deutschmann, 2021). As depicted in Equation 3-2, the WGS reaction is slightly exothermic. Hence, lower temperatures thermodynamically favor higher direct reaction conversion, yet they also decrease the reaction rate, leading to kinetic control. On the other hand, higher temperatures, as per Le Chatelier's principle, hinder the direct reaction conversion but are kinetically favored and limited by thermodynamic equilibrium (Lamb *et al.*, 2020).

To address this, the WGS process is divided into two fixed-bed adiabatic reactors operating at different temperature ranges to achieve better overall conversion (Ashok; Wai; Kawi, 2018). The inlet stream is suitably cooled to match the required operational temperature range before entering the first reactor, known as the High-Temperature Shift (HTS) reactor. This reactor operates between 350 and 550 °C and employs an iron-chromium oxide (Fe/Cr₂O₃) catalyst. Subsequently, the second reactor, called the Low-Temperature

Shift (LTS) reactor, converts the remaining CO at around 180 - 270 °C using Cu/ZnO catalysts (Alijani; Irankhah, 2013). According to literature, both reactors achieve 90% of their inlet CO conversion into H₂ (Amos, 2004).

After the Water Gas Shift (WGS) reactor, the gas mixture proceeds to an absorption column, where it comes into countercurrent contact with an appropriate solvent designed to selectively capture carbon dioxide. The efficiency of the solvent depends on factors such as temperature, pressure, type, and concentration of impurities. Physical solvents such as Dimethyl Ethers of Polyethylene Glycol (DEPG), also known as Selexol, are often used at high partial pressure of acid gas, whereas those with chemical or hybrid properties are applied in cases of low acid gas partial pressure cases or when ppm removal of acid gases are necessary (D'Alessandro; Smit; Long, 2010; Ramdin *et al.*, 2016, 2014).

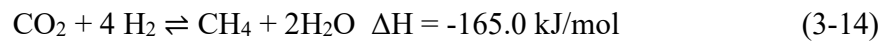
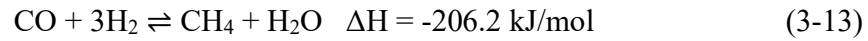
The DEPG absorption process operates within the range of -9.4 - 37,7 °C and 20 - 120 bar (Miller, 2015). Achieving significant CO₂ sequestration is challenging due to high energy requirement, a high solvent volatility, and very low selectivity (D'Alessandro; Smit; Long, 2010; Ramdin *et al.*, 2016). The size properties of DEPG molecules are valuable in ensuring low evaporative losses and viscosity, thus minimizing energy pumping costs (Tande; Seames; Benson, 2013).

After the absorption process, the solvent needs to be recovered for recycling purposes, while the released carbon dioxide is stored. The most commonly employed strategy involves the use of a stripping column, which facilitates the separation of DEPG from the dissolved gas. Carbon dioxide is released and carried away from the solvent, exiting at the top of the column, while DEPG exits at the bottom to be restored to the process (Tande; Seames; Benson, 2013).

In the context of hydrogen purification, the methanation step plays a crucial role in removing residual CO and non-absorbed CO₂ from the hydrogen stream within the column (Schmider; Maier; Deutschmann, 2021). Catalytic reactions occurring on nickel (Ni) catalysts commonly face the issue of coke formation, leading to catalyst poisoning in the active area, due to CO and CO₂. Therefore, this step is essential to reinforce the removal of carbon monoxide and carbon dioxide that were not absorbed in the column (Gardner;

Bartholomew, 1981; Schmider; Maier; Deutschmann, 2021). The methanation reactions, represented by Equations 3-13 and 3-14, are fundamental and essentially the reverse of the SMR (Equation 3-2) and SMR Global Reaction (Equation 3-3). A small quantity of remaining carbon monoxide and carbon dioxide reacts with hydrogen to form methane in a range of 250 - 350 bar (Amhamed *et al.*, 2022).

The methanation of CO₂ and CO is highly exothermic and occurs favorably at low temperatures and high pressures, typically ranging from 25 to 400 °C (Gao *et al.*, 2012; Quindimil *et al.*, 2022). Transition metals such as Ru, Rh, Pt, Pd, Ni, and Co are commonly used as catalysts for this purpose (Alrafei *et al.*, 2020; Quindimil *et al.*, 2022). Nickel, due to its advantageous properties including good activity, low cost, and high selectivity, is the primary option chosen for the process. Additionally, catalyst surface supports such as Al₂O₃, CeO₂, SiO₂, ZrO₂, TiO₂, and zeolites can be utilized. Among these, Al₂O₃ and CeO₂ are known for their cost benefits. Therefore, Ni/Al₂O₃ and Ni/CeO₂ formulations are the most widely employed catalysts for methanation in industrial applications (Quindimil *et al.*, 2022, 2018).



The reaction behavior of CO₂ methanation (Equation 3-14) related to temperature can be mathematically modeled by Equation 3-15, considering T value in Kelvin and R= 8.31 J/mol K (Quindimil *et al.*, 2022). This equation is inserted in Hysys as Keq vs temperature table, calculated for the same abscissa values as WGS process (Table 3.3).

$$K_{eq} = 137 \cdot T^{3.998} e^{\frac{158.7}{RT}} \quad (3-15)$$

Table 3.3: Equilibrium constant of CO₂ Methanation values with temperature variance.

T (K)	Keq
366.4833	2.57E+12
422.0388	4.49E+12

477.5944	7.33E+12
533.15	1.13E+13
588.7056	1.68E+13
644.2611	2.40E+13
699.8167	3.33E+13
755.3722	4.51E+13
810.9278	5.98E+13
866.4833	7.79E+13
922.0389	9.97E+13
977.5944	1.26E+14
1033.15	1.57E+14
1088.7056	1.93E+14
1144.2611	2.35E+14
1199.8167	2.84E+14
1255.3722	3.41E+14
1310.9278	4.05E+14
1366.4833	4.77E+14
1422.0389	5.60E+14
1477.5944	6.52E+14

The behavior of equilibrium constant with temperature variance for CO methanation was implemented to Hysys as a table. These data were calculated with the reverse values of the SMR's constant (Equation 3-16) in the same temperature range in Table 3.4.

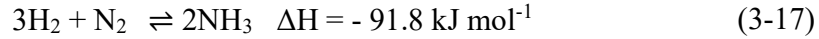
$$K_{CO-methanation} = \frac{1}{K_{SMR}} \quad (3-16)$$

Table 3.4: Equilibrium constant of CO Methanation values with temperature variance.

T (°C)	K	K⁻¹
93.3333	7.81E-19	1.28E+18
148.8888	6.84E-15	1.46E+14
204.4444	7.79E-12	1.28E+11
260	2.17E-09	4.60E+08
315.5556	2.19E-07	4.57E+06
371.1111	1.02E-05	9.77E+04
426.6667	2.66E-04	3.76E+03
482.2222	4.34E-03	2.31E+02
537.7778	4.90E-02	2.04E+01
593.3333	0.4098	2.44
648.8889	2.679	3.73E-01
704.4444	14.26	7.01E-02
760	63.43	1.58E-02
815.5556	242.6	4.12E-03
871.1111	816.6	1.22E-03
926.6667	2464	4.06E-04
982.2222	6755	1.48E-04
1037.7778	17010	5.88E-05
1093.3333	39670	2.52E-05
1148.8889	86640	1.15E-05
1204.4444	178400	5.61E-06

Subsequently, the hydrogen and nitrogen mixture is ultimately directed to the ammonia synthesis reactor, commonly considered the core of the ammonia production unit. This reactor is characterized by an exothermic reaction that yields a stable product without

side reactions. This reactor operates at 300 - 500 °C and 150 - 200 bar conditions (Moura, Igor de Paulo, 2021), reacting nitrogen and hydrogen to produce ammonia (Equation 3-17) (Rouwenhorst, Kevin; Ham; Lefferts, 2021). This reaction is heterogeneous catalyzed by iron-based surfaces and keep up with a cooling system to favor better conversion (Khademi; Sabbaghi, 2017).



The most commonly reported equation model in the literature regarding the kinetics of ammonia production is the Temkin-Pyzhev model (Equation 3-18) (Khademi; Sabbaghi, 2017).

$$R_{\text{NH}_3} = 2k \left[K_a^2 a_{\text{N}_2} \left(\frac{a_{\text{H}_2}^3}{a_{\text{NH}_3}^2} \right)^\alpha - \left(\frac{a_{\text{NH}_3}^2}{a_{\text{H}_2}^3} \right)^{1-\alpha} \right] \quad (3-18)$$

Where k represents the kinetic constant of the reverse reaction, K_a stands for the equilibrium constant, a_i represents the activity of the specific component ($i = \text{H}_2, \text{N}_2$, and NH_3), and α is a constant that varies between 0.4 and 0.75. For iron-based catalysts, such as one consisting of 95% Fe, 2% Al_2O_3 , 1.5% K_2O , and 1.5% CaO , α is assumed to be 0.5 (Satimurty, 2014). In this scenario, by simplifying the equation to ideal gas conditions, it is possible to obtain the ammonia rate using only the kinetic parameters of the Arrhenius direct and reverse reactions as shown in the Equation 3-19.

$$R_{\text{NH}_3} = f \left(k_{1,0} \exp \left(\frac{-E_{a1}}{RT} \right) \frac{P_{\text{N}_2} P_{\text{H}_2}^{1.5}}{P_{\text{NH}_3}} - k_{2,0} \exp \left(\frac{-E_{a2}}{RT} \right) \frac{P_{\text{NH}_3}}{P_{\text{H}_2}^{1.5}} \right) \quad (3-19)$$

Where $k_{1,0}$ and $k_{2,0}$ are the pre-exponential factors of Arrhenius, and E_{a1} and E_{a2} are the activation energies respective to the forward and reverse reactions. The parameter R represents the universal gas constant, T is the temperature within the reactor, P_i is the partial pressure of each component ($i = \text{H}_2, \text{N}_2$, and NH_3), and f is the catalyst activity factor (Murase;

Roberts; Converse, 1970).

3.3

Implementation of HB in Aspen Hysys

Initially, the steam/methane ratio is set to 3.34 based on Devos considerations (De Deken; Devos; Froment, 1982). While some side reactions occur, in this work, the First Reformer simulation simplifies the interaction between methane and steam as a single reaction, namely Steam Methane Reforming (SMR). This approach is crucial for optimizing computational efficiency and ensuring convergence in Aspen Hysys, enabling the exploration of different conditions in the sensitivity analysis section.

In the context of this study, it is important to note that the natural gas feed stream exclusively comprises methane. Devos (1982) has asserted that the small fraction of other light hydrocarbons present in natural gas undergoes rapid decomposition, resulting in the production of methane, hydrogen, and carbon dioxide. Consequently, employing a feed stream that solely consists of methane serves as a suitable approximation and facilitates the simulation process (De Deken; Devos; Froment, 1982).

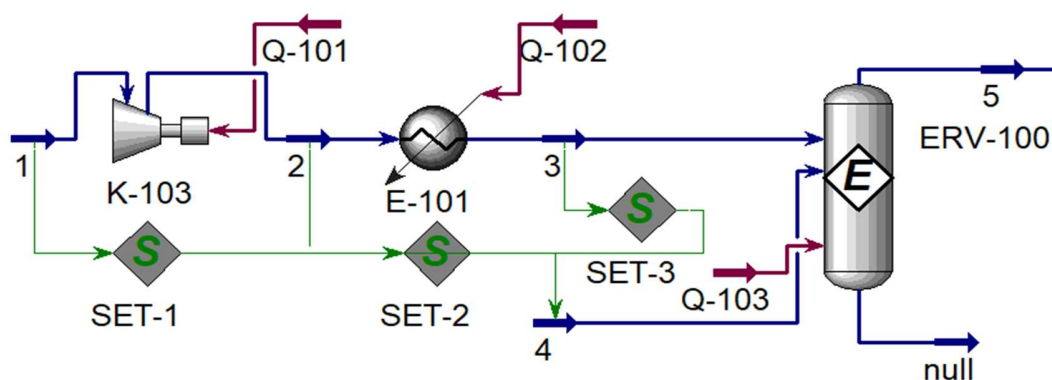


Figure 3.3: First Reformer Unit.

For the optimization of independent variables of the first reformer, a compressor and a heater were added to calculate energy consumption due to variation in the vessel's pressure and temperature. The reactor inlet temperature is considered constant at 520 °C (De Deken; Devos; Froment, 1982). To perform total energy consumption calculation, the values of power Q-101, Q-102, Q-103 were summed (Figure 3.3). A case study done in the software

recreated 2091 scenarios of different combinations of reactor's temperature and pressure, writing down results to create a sensibility surface, these parameters were worked in range introduced in Table 3.5 and the constant conditional settings in Table 3.6.

Table 3.5: SMR conditions range in sensibility analysis.

	Minimum	Maximum
Temperature (°C)	800	1000
Pressure (bar)	20	35

Table 3.6: SMR constant conditional settings.

Methane Flow Rate (kmol/h)	269.40
Inlet Temperature (°C)	520.00
Steam/methane ratio	3.34

The operating section of the second reformer was defined from the outlet stream of the first reformer (stream 5) to the outlet of the second reformer (stream 12), as shown in Figure 3.4.

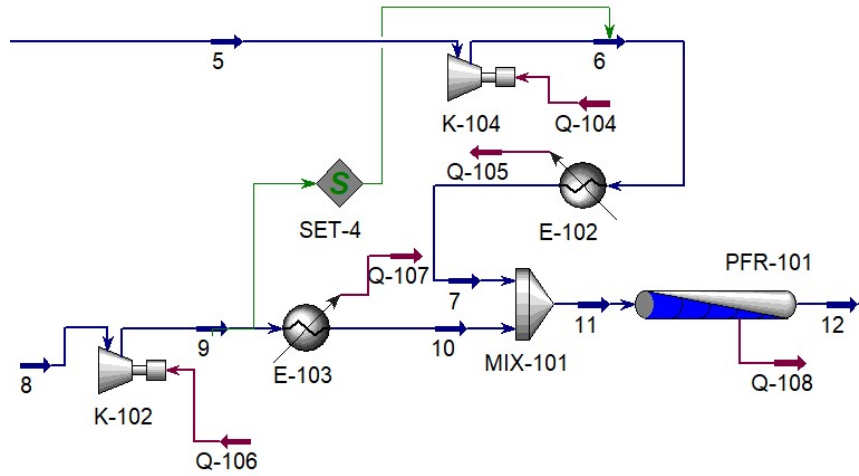


Figure 3.4: Second Reformer Unit.

Stream 8 represents the air inlet, being pressurized in the same manner as the other parallel streams according to the reforming operating pressure. The ideal amount of air is the quantity sufficient to achieve stoichiometric proportions of nitrogen and hydrogen. The inlet temperatures were kept constant based on literature (Table 3.7) values (AL-Dhfeery; Jassem, 2012). For sensibility analysis, the pressures of stream 9 and 6 varied equally between 30 - 50 bar, while the reactor temperature ranged from 950 °C to 1050 °C.

Table 3.7: Constant feed temperatures of Second Reformer.

Feed Stream	Temperature (°C)
Air	550
Unreacted Methane	795

In our study, the Jones-Lindstedt model implemented in the PFR-101, that is filled by a series of slender and elongated tubes with a diameter of 0.1 m and a length of 10 m to simulate the secondary reforming reactions (Jain *et al.*, 2021). The outlet of the first reformer and the air will both be supplied through the initial orifice.

After the second reformer, the pressure and temperature of stream 12 are adjusted to meet the operational requirements of the High-Temperature Water Gas Shift (HT-WGS) reactor (Figure 3.5). Subsequently, the outlet stream from the HT-WGS reactor undergoes cooling before entering the Low-Temperature Water Gas Shift (LT-WGS) reactor. The hydrogen production unit of the HB plant is thus completed at the outlet of stream 17.

The shift reactors were also simulated in equilibrium reactors that use only the WGS reaction (Equation 3-2), modeled by the thermodynamic parameters presented earlier (Figure 3.2). The operating pressure was set constant at 30 bar, while the temperature of the two reactors operated within their respective ranges, 350 – 550 °C and 180 – 270 °C. Both stages were operated in an adiabatic process, and due to the slightly exothermic nature of the reaction, the outlet temperature is higher than the inlet temperature. Therefore, the decision was made to adjust the inlet temperatures to ensure that the temperature-dependent variables of the reactors fell within the ranges reported in the literature. The manipulated values were systematically varied through screening, with increments of 2.5 °C aiming to achieve results within the operating temperature range of their respective reactors.

The WGS reactors are preceded by a compressor and a heat exchanger to adjust the inlet temperatures and pressures. Similarly, between the HTS and LTS reactors, there is another cooler that adjusts the outlet temperature of the first reactor to the inlet temperature of the second one.

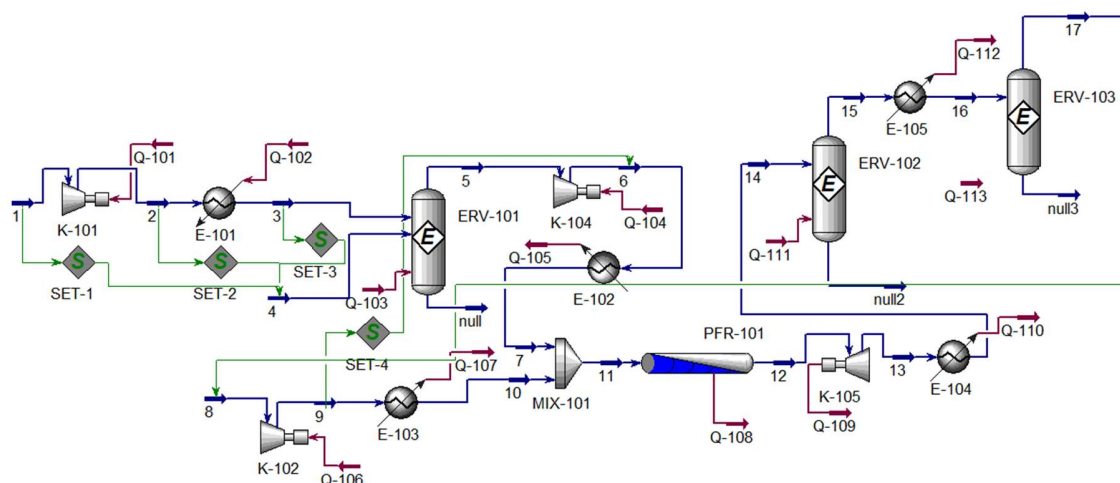


Figure 3.5: HB hydrogen production unit.

Following the hydrogen production unit, the produced gas stream proceeds to the CO₂ capture section (Figure 3.6). In this stage, the gas enters a countercurrent absorption column, where it contacts an appropriate solvent (DEPG, stream 19) to absorb carbon dioxide, while the unabsorbed gas exits through stream 23 and is recycled back into the system. The DEPG solvent, now rich in CO₂, proceeds to a solvent regeneration column and is recycled back to the absorption column. Streams 29, 30, and 31 represent the outlets for CO₂, steam, and regenerated DEPG, respectively. A small amount of water exits via stream 34 to prevent accumulation within the process. Subsequently, DEPG undergoes a purge step in equipment X-101.

The CO₂ capture process was simulated using a 20-stage absorption column, testing specific pressure and temperature conditions within the ranges reviewed in the literature (Sulaiman; Matloub; Shareef, 2018). To achieve this, a compressor and a heat exchanger were employed prior to the process, and the operation conditions of 25 °C and 120 bar, 37.77 °C and 120 bar, and 25 °C and 30 bar were tested for carbon dioxide capture from the HB plant. The amount of solvent in each scenario was adjusted using Aspen Hysys through iterations to determine the minimum molar flows required to achieve 95% carbon dioxide absorption using the secant method.

The absorption process is followed by a flash tank operating at the same outlet temperature. The pressure is adjusted through simulator iterations using the same tool as before to recycle the bottom H_2 outlet back to the column. This minimizes the hydrogen gas content in the liquid phase sent to the stripper with a maximum allowance of 1.5 kmol/h. Therefore, for each scenario, the flash tank operated at an appropriate pressure calculated by the simulator, resulting in a different bottom outlet reflux. Subsequently, the stripping process separates the CO_2 from the solvent counter currently using water vapor, while the concentrated DEPG exits at the bottom and is recycled to the beginning of the capture process. The stripper column operated with a pressure ranging from 200 - 300 kPa between the initial and final trays, with a 100 kPa difference for the condenser and reboiler in all three situations. Meanwhile, the temperature of each stage was also calculated by Aspen Hysys for the three different scenarios.

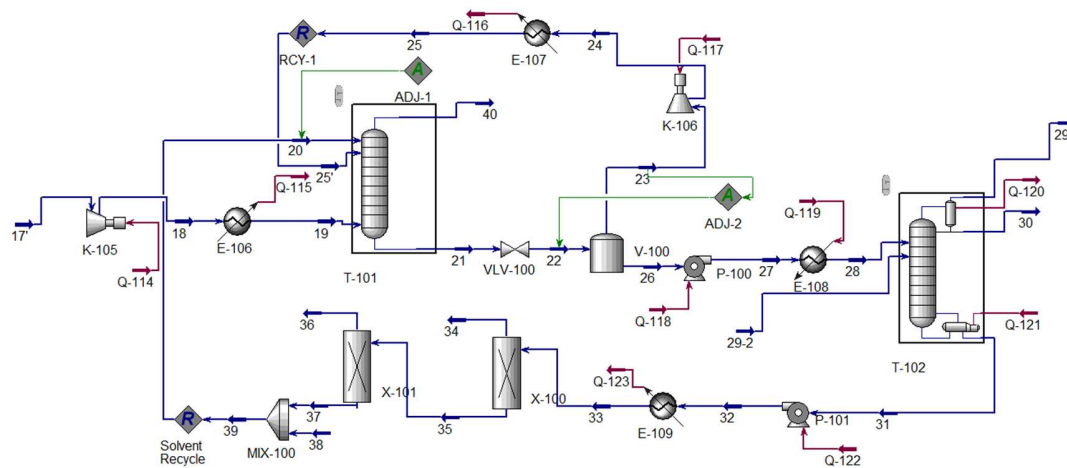


Figure 3.6: HB CO_2 capture unit.

The ammonia synthesis unit in the HB plant utilizes methanation through reactor ERV-104 followed by a dehydration column X-103, as depicted in Figure 3.7. Considering that the methanation process operates in the reverse direction of SMR, some of the hydrogen is lost. Therefore, it is necessary to update the value of atmospheric air entering the second reformer, iteratively rebalancing until the amount of nitrogen inserted corresponds to one-third of the amount of hydrogen remaining after methanation. This iteration was initially done manually by manipulating the air flow, and as the value approached, the secant method

was used through the simulator's adjustment tool to minimize the error. Therefore, all results recorded after the second reformer already account for the slight hydrogen loss during methanation, thus reflecting the optimal final conditions regarding air flow rate and the necessary quantity of DEPG, for example.

The ammonia synthesis reactor was modeled as a Plug Flow Reactor (PFR), incorporating appropriate kinetic parameters. Operating conditions of pressure and temperature were systematically varied within ranges of 150–300 bar and 300–500 °C, respectively. All kinetic data required for simulating the iron-based catalyst behavior in Aspen Hysys are detailed in Table 3.8.

Additionally, a sensitivity analysis of the ammonia synthesis reactor was conducted by evaluating different reactor configurations to identify optimal operational conditions and strategies aimed at maximizing productivity per unit of catalyst bed volume. As an example, Figure 3.7 depicts the autothermal fixed-bed reactor configuration employed in conjunction with ammonia looping. The following subsection of this chapter will describe each configuration tested for ammonia production.

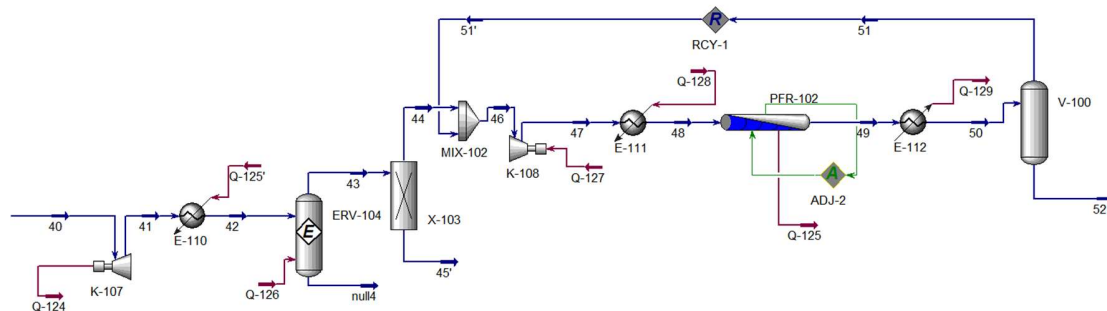


Figure 3.7: HB methanation and ammonia synthesis unit.

Table 3.8: Ammonia Reactor Kinects Parameters used in Simulation.

ϵ	0.42	-	Void fraction
$C_{p,c}$	1100	$\frac{J}{kg\ K}$	Specific heat capacity of catalyst.

pc	2200	$\frac{kg}{m^3}$	Packing density of catalyst.
f	1	-	Catalyst activity factor.
E_{a1}	20800	$\frac{kcal}{kmol}$	Activation Energy of direct reaction per kmol of N ₂ .
E_{a2}	47,400	$\frac{kcal}{kmol}$	Activation Energy of reverse reaction per kmol of N ₂ .
k_{0,1}	1.78954 x 10 ⁴	$\frac{kmol}{m^3 h atm^{1.5}}$	Pre-exponential Arrhenius factor of direct reaction.
k_{0,2}	2.5714 x 10 ¹⁶	$\frac{kmol atm^{0.5}}{m^3 h}$	Pre-exponential Arrhenius factor of reverse reaction.

3.3.1

Adiabatic PFR single bed reactor

The first case to be simulated was using an adiabatic PFR reactor with a single fixed bed with a diameter of 1.1 m. As the reaction is exothermic, the inlet temperature was strategically set at 300 °C, which is the lower limit of the literature range. The pressure was then varied at 150, 200, and 300 bar to analyze the behavior of conversion, temperature, and rates along the length of the reactor in each scenario. At the end of each stage, the optimal reactor volumes were determined for each pressure scenario and subsequently used as inputs for evaluating the following strategy. The product and the remaining reagents are separated through a flash tank, where ammonia is obtained at bottom of the flash vessel.

The subsequent phase involves incorporating an ammonia loop system, where the unreacted components exiting the flash tank's top are entirely refluxed back to the bed reactor. As the overall feed rate increases, it becomes crucial to recalibrate the reactor volume iteratively using the secant method. The production of the targeted theoretical ammonia yield (100%) is ensured while upholding the reactor's maximum conversion rate as observed in the loop-free scenario. This is possible by determining the number of cycles of reagents inside the loop and adjusting the reactor volume accordingly using this simulation tool.

3.3.2

Autothermal PFR single bed reactor

Next, the same was done with the autothermal reactor, where the reaction does not occur adiabatically in a PFR, as there are cooling tubes to control the temperature inside the reactor. Firstly, the behavior of conversion and useful heat released from the reactor was analyzed for different pressure values and inlet and outlet temperatures to find the best operating combination for two different cases: a 6 m³ volume reactor and a 30 m³ volume reactor. The first is related to the same value used in the adiabatic bed reactor for comparison purposes, and the second is the approximate value at which the reactor begins to reach its maximum conversion. Additionally, a looping was added, readjusting the reactor volume and new flow rates inside loop using the same iterative resource of the simulator.

3.3.3

Adiabatic multiple beds intercooling system

Finally, intercalated beds with a cooling system were simulated. The volumes were initially adjusted by extrapolating a large volume for each of them, and through the simulator's performance tab, it was possible to determine at which volume the conversion of each reactor ceased to increase significantly. These values were noted and inputted from the first to the last reactor, set up in this order. Consequently, each of the reactors was resized in its proper sequence to the minimum volume required to achieve a conversion very close to the set value.

To increase conversion, the nitrogen strategy was tested, first determining the amount of excess nitrogen for conversion, and then readjusting the volume values. The result was compared to the stoichiometric case through a graph showing the effect of the required volume reduction.

3.4

LAC Steps, Reactions and Modeling

The LAC process features most of the same simulations, equations, and reactions as the HB process. The main distinction lies in the absence of a secondary reformer, necessitating the complete conversion of methane during the SMR stage, followed by the WGS reaction. It is worth mentioning that, in the LAC process, the Water Gas Shift reaction may be carried out in a single reactor configuration, known as the Medium Temperature Shift (MTS) reactor. Typically operating within the temperature range of 280 °C to 360 °C, this reactor may employ catalysts such as Platinum (Pt), Nickel (Ni), Gold (Au), Palladium (Pd), and Rhodium (Rh), supported on materials like CeO₂, TiO₂, ZrO₂, Al₂O₃, CeO₂-ZrO₂, and MgO (Alijani; Irankhah, 2013). This alternative is particularly suited to LAC plants, as the subsequent stages involve hydrogen purification through Pressure Swing Adsorption (PSA) technology, which ensures the effective removal of carbon monoxide and other residual gases.

The pressure swing adsorption (PSA) process was first developed in 1966 and has since become a crucial purification step for hydrogen production. Following the absorption system, it is essential to remove residual carbon dioxide and moisture. While various strategies like solvent extraction and membrane contactors can be employed, PSA is particularly well-suited for ammonia production in the LAC method. By operating at ambient temperatures, PSA ensures 99.999% pure hydrogen, thus preventing catalyst poisoning from carbon monoxide and other contaminants.

PSA is a unit operation that separates gas mixtures into adsorbed and non-adsorbed components based on their specific surface interactions. The choice of adsorbent material is critical for achieving optimal selectivity, taking into account factors such as polarity, pore size, kinetic energy (temperature), mean free path (pressure) of the adsorbate molecules, and the temperature of the adsorbent itself.

The adsorbent material is added into the pressure vessels, forming a fixed bed with a certain height. The gas percolates through this fixed bed for the separation of the desired

contaminants. The alternating adsorption and desorption phenomena take place in two separate pressure vessels, with one vessel adsorbing while the other desorbs, controlled by pressure changes. Typically, unwanted impurities are trapped in the pores surface and the non-adsorbed gas (hydrogen) pass through. Once adsorption is complete in one vessel, depressurization begins to regenerate the bed, part of product arrested in voids are 86% recovered and together with impurities released, it is used to pressurize the other vessel, continuing the cycle.

Commonly used materials for selective removal of impurities like carbon monoxide, carbon dioxide, methane, and other hydrocarbons include activated carbon, superactivated carbon, 5A zeolite molecular sieves, and silica gel. Modern technology enables the production of high-quality purified hydrogen at a rate of approximately $100,000 \text{ Nm}^3 \text{ h}^{-1}$, with minimal product loss, utilizing up to 12 beds with higher adsorption capacity (Sharma, 2009).

The Air Separation Unit (ASU) generates high-purity nitrogen, which serves as the feedstock for the ammonia synthesis reactor in LAC plants. Atmospheric air undergoes filtration, compression, and purification to remove dust and other impurities. According to the literature, certain ASU processes operate similarly to Pressure Swing Adsorption (PSA) systems, utilizing appropriate adsorbents for gas separation (Sharma, 2009). This separation occurs at low temperatures, ensuring high energy efficiency, reduced operating costs, and simplified processing (Deng *et al.*, 2022). PSA technology can also be used for the separation of other gas mixtures, such as nitrogen and oxygen. By employing an activated carbon molecular sieve (CSM), oxygen is selectively adsorbed, allowing nitrogen to pass through and be collected for use (Reynolds; Ebner; Ritter, 2006).

In the LAC process, nitrogen is introduced directly into the ammonia reactor. This work aims to evaluate the impact of nitrogen excess by applying this strategy to the reactor configurations previously tested for the HB plant.

3.5

Implementation of LAC

The simulation begins in the hydrogen production unit, where stream 1 is fed with methane gas, replicating all numerical values and parameters specified in the SMR section of the HB plant (Figure 3.8). The pressure and temperature conditions of the SMR stage are set based on the maximum methane conversion point determined from the response surface analysis, ensuring that SMR, as the only methane reforming stage, achieves its highest efficiency.

A key distinction in the LAC process is that, following the initial reforming step, stream 5 undergoes temperature and pressure adjustments to be directly fed into the High-Temperature Water Gas Shift (HT-WGS) reactor. Subsequently, similar to the HB process, the outlet stream from the HT-WGS reactor is cooled to an appropriate temperature before entering the Low-Temperature Water Gas Shift (LT-WGS) reactor. The optimal temperature conditions for these reactors are derived from the results of the HB plant and subsequently replicated in the LAC plant, whereas the pressure condition remains unchanged.

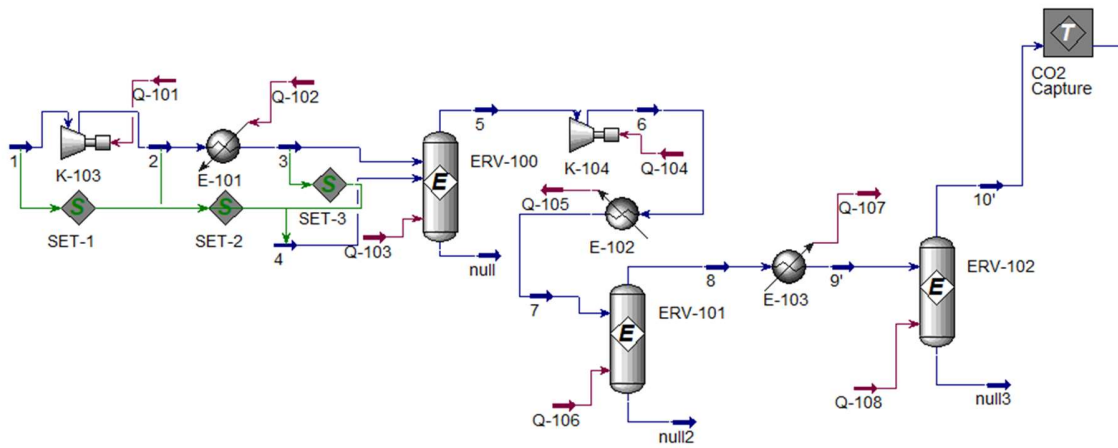


Figure 3.8: LAC hydrogen production unit.

Following the hydrogen production stage, the gas stream is directed to the CO₂ capture unit, whose operation is analogous to that of the corresponding section in the HB

plant. The only notable distinctions lie in the numbering of streams and equipment, as illustrated in Figure 3.9. Meanwhile, the pressure and temperature conditions applied in the absorption column are replicated from the optimal parameters previously determined in the HB plant.

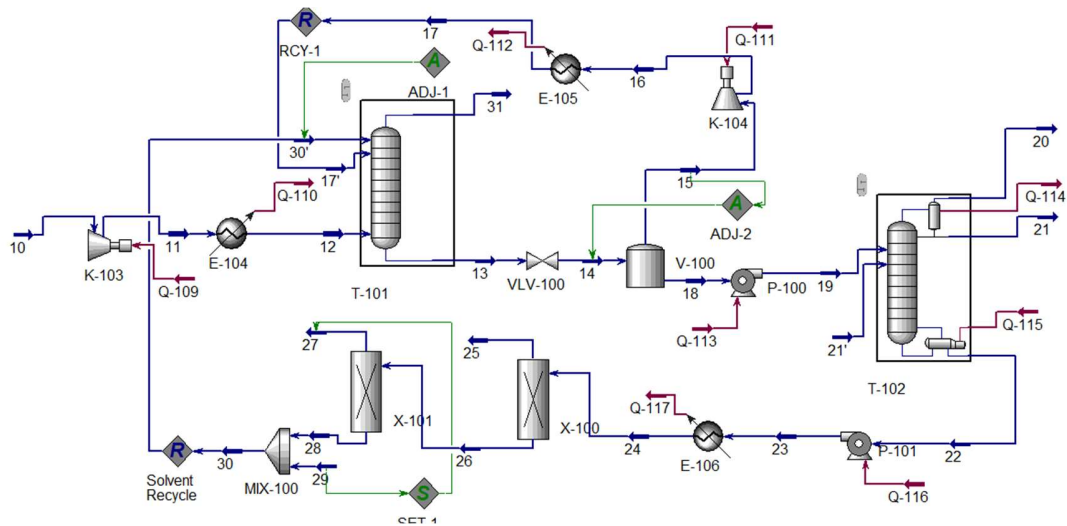


Figure 3.9: LAC hydrogen production unit.

The ammonia synthesis unit of the LAC plant features a Pressure Swing Adsorption (PSA) system, represented by equipment X-103, for hydrogen purification, which replaces the methanation and dehydration processes utilized in the HB plant. Additionally, the LAC plant receives nitrogen streams 41 and 42 from an Air Separation Unit (ASU), which are mixed directly with the purified hydrogen stream prior to entering the ammonia synthesis reactor, as illustrated in Figure 3.10.

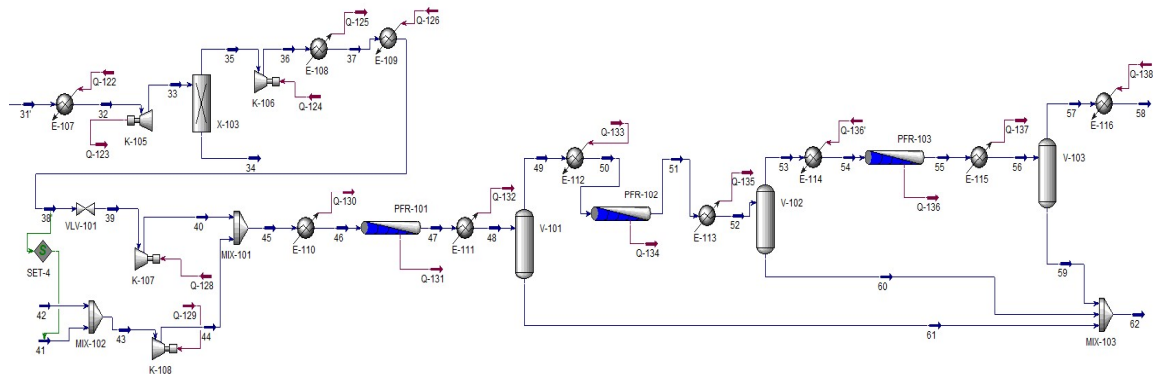


Figure 3.10: LAC PSA and ammonia synthesis unit.

3.5.1

Nitrogen Excess Strategy

In this study, the excess nitrogen strategy will be exclusively implemented for the LAC plant. Therefore, it is necessary to readjust the chosen reactor for the procedure according to the change in the feed flow, since the LAC produced more hydrogen than the HB plant. Firstly, the 30 m³ single-bed autothermal reactor was readjusted to 34.57 m³ to maintain constant conversion compared to the previous study. This was also done for the looping system reactor, where 56.39 m³ became 66.88 m³. In the case without looping, the effect of excess nitrogen was studied by increasing its volume to maintain the space time in the reactor constant as the excess reagent quantity grows. Thus, the ideal excess quantity was found.

Next, two additional intercalated reactor beds were incorporated into the system, separated by a flash separator. Flash separators are placed between these reactors to remove the product, thereby increasing the conversion of the following reactor. The sum of the volumes of the three beds was chosen to be 66.88 m³ in order to compare the results based on the same catalyst volume used in the autothermal reactor with looping. Initially, the volumes are directly based on the proportion of their respective feeds. After that, the volumes

among the three reactors are redistributed to find the best combination. The optimizer tool was used to manipulate the reactor volumes to minimize the ratio between the total volume and the amount of ammonia produced. After finding the best combination, a reevaluation of the nitrogen excess value is carried out while maintaining the volumes.

3.6

Energy integration

Once each plant's simulation has been completed under the optimal operating conditions (temperature, pressure, component flow rates, and reactor configurations) defined for each stage, the energy analysis tools provided by Aspen Hysys software are utilized. Through this analysis, Aspen Hysys determines the appropriate utility type for each energy stream and outlines the optimal strategy for energy integration. The utility streams are detailed based on Aspen Hysys's own manual, as shown in Table 3.9. For both plants, a comprehensive description of the energy integration process, including details on all utility streams employed, is provided in the appendices of this study.

Table 3.9: Utilities stream of Aspen Hysys used for energetic analysis.

Utility Stream	Typical Substance or Medium Employed
Fired Heat (1000, 2000)	Natural gas combustion gases
Very High Temperature	Combustion gases from methane reforming
Hot Oil	Thermal oil (Dowtherm, Therminol)
Refrigerant 1	Ammonia (NH ₃)
Refrigerant 2	Propylene or Ethylene (typical for moderate cooling needs)
Refrigerant 3	Ethylene or synthetic refrigerants (lower temperature ranges)
Refrigerant 4	Propane or other synthetic refrigerants (deep refrigeration)
Cooling Water	Water (from cooling towers)
Air	Ambient air
Very Low Temperature	Refrigerant (Ammonia or synthetic refrigerants for deep cooling)
LP Steam Generation	Low-pressure saturated steam (~3-5 bar)
MP Steam Generation	Medium-pressure saturated steam (~10-20 bar)
HP Steam Generation	High-pressure superheated steam (above ~40 bar)
Utility Stream	Typical Substance or Medium Employed

4.

Results and Discussion

4.1

First reformer

Analyzing the resulted sensibility surface, it is possible to see the performance of SMR reaction according to the operation conditions. Increasing the temperature of the first reformer leads to an increase in methane conversion values until gradually, this acceleration in growth diminishes, representing a convergence behavior, as reaching values close to 100% becomes more difficult. Meanwhile, the reactor's pressurization condition exhibits an inverse behavior. As pressure increases, the equilibrium of the SMR reaction is shifted in favor of the reactants. This was expected due to the principle of Le Chatelier's law, as the products side of the SMR reaction has a higher number of moles (Figure 4.1).

For the hydrogen flow rate, a similar behavior is observed. However, it is important to mention that hydrogen production is associated with the simultaneous progress of the combination of two reactions present in the reactor, the SMR and WGS, namely, the Global SMR reaction (Equation 3-3). While the methane conversion graph (Figure 4.2) solely relates to the progress of the SMR reaction (Equation 3-1).

It is important to note that although the WGS reaction is involved in this reactor, the operating conditions are not suitable for complete conversion, as can be observed in the Figure. This means that the conversion of CO to hydrogen is only complete after the subsequent steps. On the other hand, the total heat supplied to the section increases drastically with temperature, as higher SMR conversion values are achieved under these conditions, requiring the supply of highly endothermic enthalpy of the main reaction (Figure 4.3).

Finally, the optimal conditions for each plant were defined. Since the HB process involves a subsequent reactor that continues the steam reforming reaction combined with partial oxidation, it was understood that it was not necessary to opt for the condition with the highest yield, but rather the one with a better hydrogen production-to-energy ratio (Equation 4-1).

$$\frac{\text{Production}}{\text{Energy spent}} = \frac{M_{H_2}}{E_{total}} \quad (4-1)$$

The highest value found, considering a calculation basis of one hour, was 7.85E-5 kmol (MWh)⁻¹ under lower pressure and temperature conditions. Theoretically, the lower temperature not only demands a lower enthalpic energy value due to the low SMR conversion but also because the WGS reaction conversion would be higher and slightly exothermic, it provides more energy to the system, reducing the electrical energy consumption. On the other hand, the criterion used for the LAC plant was the highest H₂ conversion since it would be the only reformer, and the objective was to minimize the losses of unconverted methane. Table 4.1 lists these established conditions.

Table 4.1: First Reformer Conditions.

Process	Temperature (° C)	Pressure (bar)	Conversion (%)	Heat (MWh)
HB	800	20	76.73	18.2
LAC	1000	20	98.76	25.6

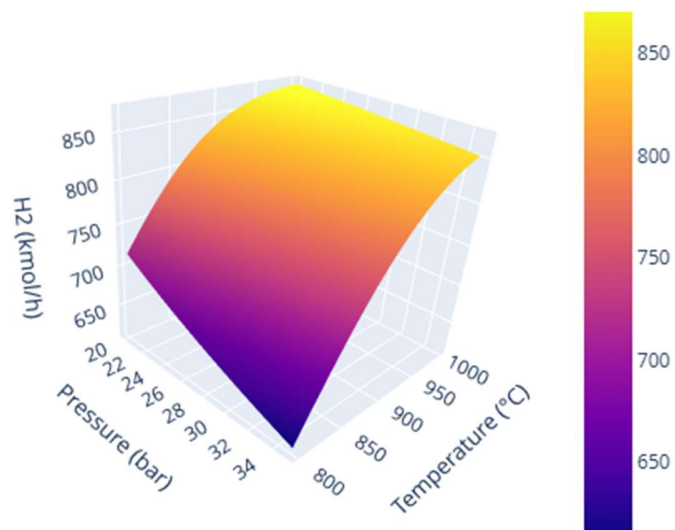


Figure 4.1: Hydrogen flow rate produced in the first reformer.

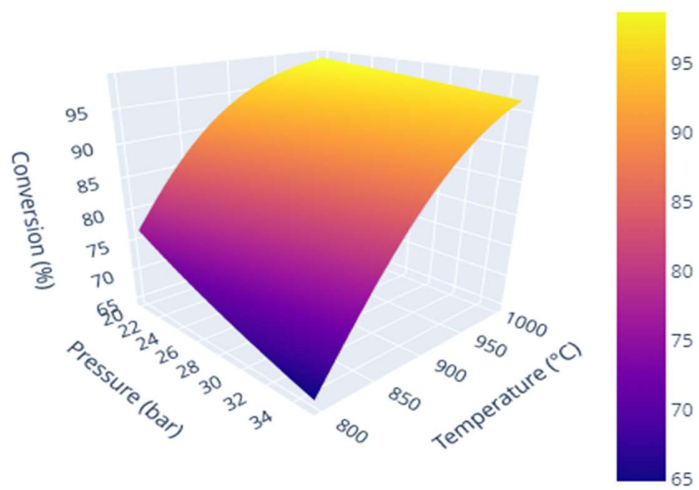


Figure 4.2: SMR conversion.

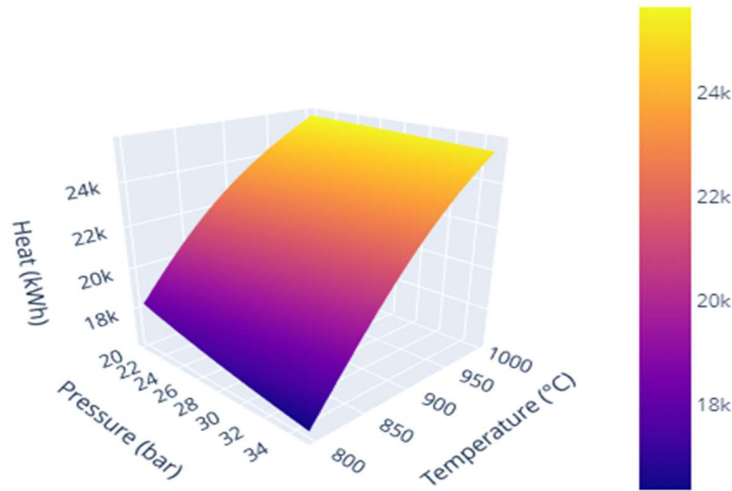


Figure 4.3: First reformer section's total Power required.

4.2

Second Reformer

The air inlet flow was adjusted to 390 kmol/h to guarantee 304 kmol/h of nitrogen stoichiometrically balanced to approximately 919 kmol/h of H_2 at the end of hydrogen production unit. Based on the temperature and pressure sensitivity study, the second reformer exhibited different results in the conversions of its four different reactions under different scenarios (Figures 4.4, 4.5, 4.6 and 4.7). It can be observed that the partial oxidation of methane reaction was more sensitive to pressure variations, while the others were more sensitive to temperature variation.

It was observed that the POM and SMR reactions are compensatory reactions based on variations in operational conditions. At higher temperatures and pressures, the reactor is conditioned to operate with high steam reforming conversions. Conversely, in lower temperature and pressure conditions, partial oxidation of methane is predominant. Although

the first reaction is theoretically desired to be predominant as it produces more hydrogen, high temperature conditions favor the reverse direction of the WGS reaction and hydrogen combustion. Starting from 1005 °C, the water-gas shift (WGS) reaction proceeds in the reverse direction, which mathematically would result in negative conversion values. However, from a physical standpoint, negative conversion values have no practical meaning. For this reason, Figure 4.6 limits the scenarios depicted in the graph to temperatures above 1005 °C.

Finally, the simulated amount of hydrogen produced was found to be sensitive only to the reactor's operating temperature, varying from 741.78 to 754.65 kmol/h (Figure 4.8) as temperature increased. However, pressure had no significant influence on hydrogen production, as it shifts equilibrium solely toward hydrogen combustion, consuming hydrogen, while the other reactions rebalance its quantity. Similarly, carbon monoxide exhibited analogous but inverse behavior, with production increasing as temperature decreased, ranging from 165.81 to 178.69 kmol/h (Figure 4.9). Therefore, it can be concluded that as hydrogen production increases, the amount of carbon monoxide decreases. Since the next stage does not convert 100% of the carbon monoxide, it was considered that the optimized condition would be the one with the highest immediate hydrogen production. Considering the quantity and the scenario with the lowest source of hot heat to minimize losses, the chosen operating conditions for the second reformer were 950 °C and 30 bar.

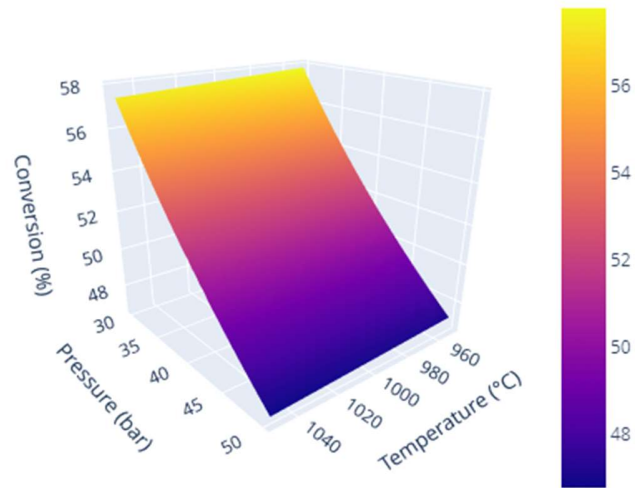


Figure 4.4: Second reformer POM conversion.

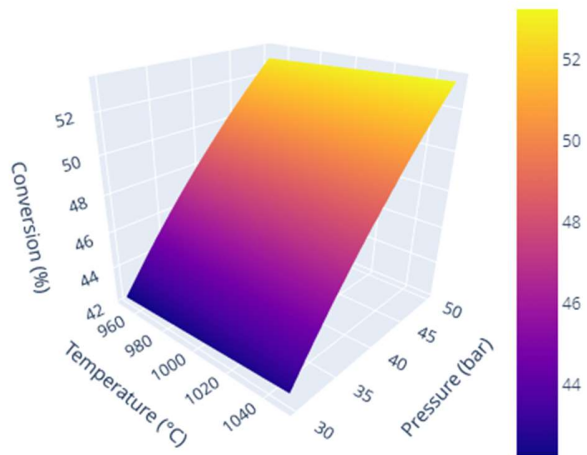


Figure 4.5: Second reformer SMR conversion.

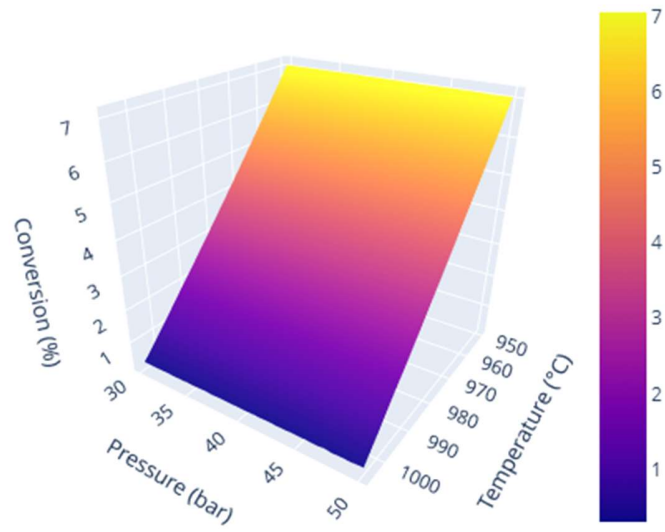


Figure 4.6: Second Reformer WGS conversion.

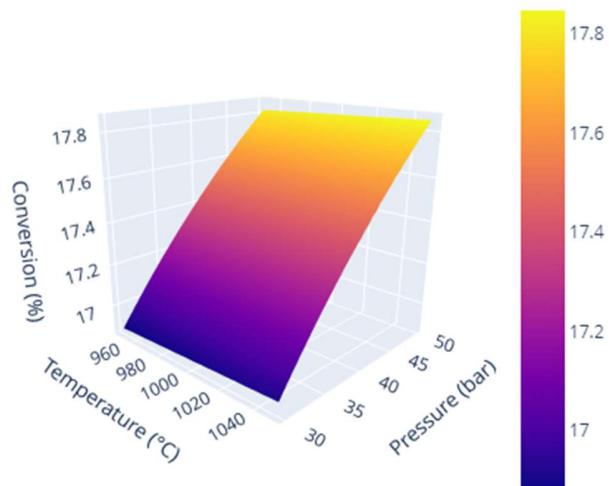


Figure 4.7: Second Reformer H₂ combustion.

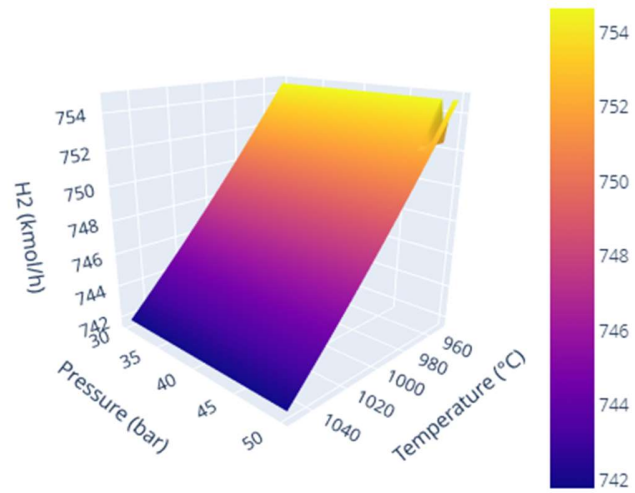


Figure 4.8: H₂ flow rate after second reformer.

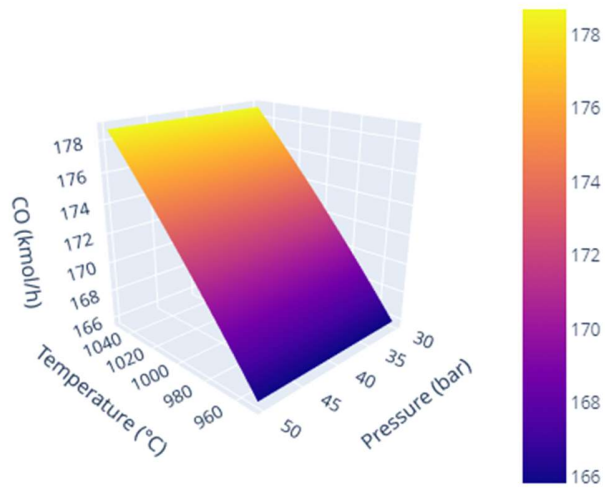


Figure 4.9: CO flow rate after second reformer.

4.3

WGS

From the tested operating conditions, the shift reactor stages exhibited high conversions and achieved results around 90%, consistent with the literature (Amos, 2004). Due to thermodynamic considerations, the decreasing behavior of conversion with temperature was observed in both reactors of the two plants. The exothermic nature of the reaction allows for higher conversion values at lower temperatures (Figures 4.10 and 4.11). The lower operating temperatures were obtained based on the tabulated feed temperatures (Table 4.2). It can be observed that the LAC requires a lower inlet temperature compared to the conventional plant. This is due to the difference in stream compositions. In the HB plant, the second reformer already introduces nitrogen gas into the process as an inert component, which absorbs a portion of the heat released by the reaction. Therefore, it requires hotter streams at the inlet of the reactors. These energy considerations will be crucial in comparing the two plants at the end.

Table 4.2: WGS feed and outlet temperatures of HB and LAC plants.

	Feed Temperature (°C)	Outlet temperature (°C)
HT	270	352.16
HT-LAC	240	351.68
LT	170	181.38
LT-LAC	162.5	182.45

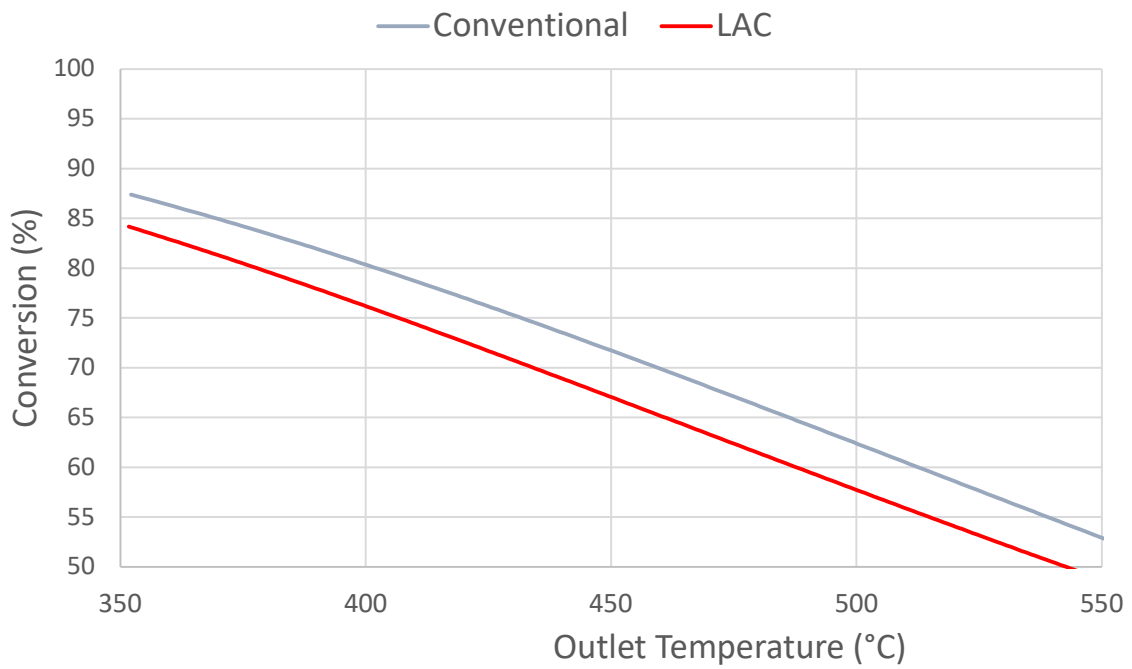


Figure 4.10: HT-WGS relation between conversion and outlet temperature.

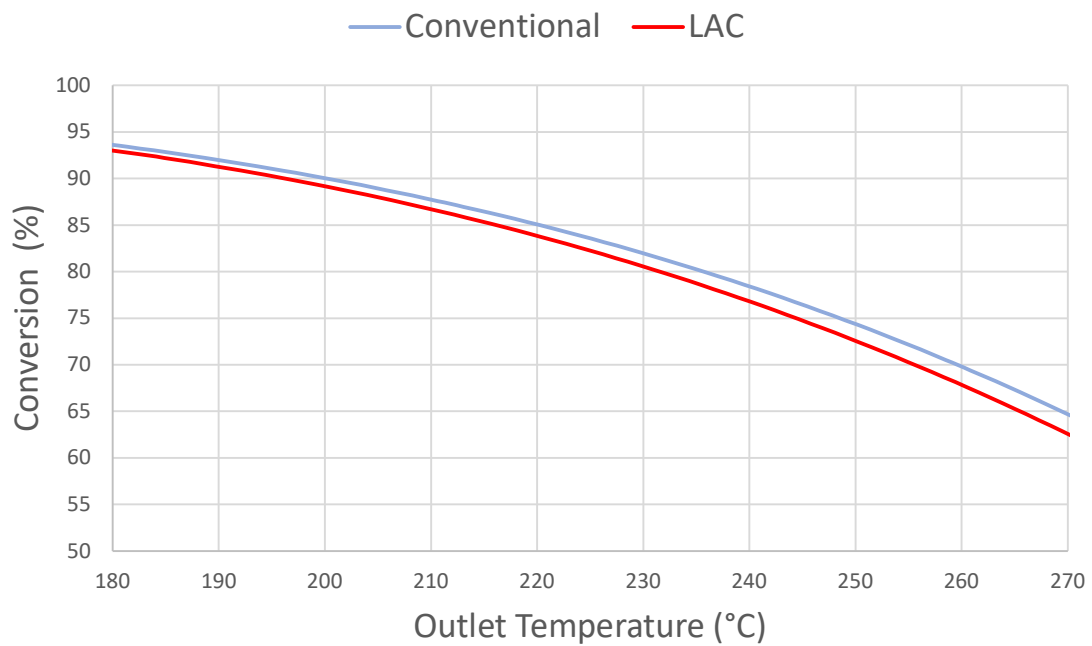


Figure 4.11: LT-WGS relation between conversion and outlet.

4.4

CO₂ capture

The results obtained from the capture unit tests in the HB plant were recorded. After determining the optimal temperature and pressure conditions and replicating them for the capture section of the LAC plant, complete tables regarding the capture performance (Table 4.3) and energy consumption (Table 4.4) were obtained. Finally, it is possible to critically analyze the relationship between CO₂ recovery, desired solvent quantity, and energy consumed during the process.

With these results, it is possible to evaluate how high-pressure values and low temperature values significantly decrease the required solvent quantity. High pressure facilitates gas displacement into the liquid phase, while temperature, representing molecular agitation, poses difficulty for gas to attach to the liquid phase, decreasing its solubility. An important point to note is that the amount of absorbed carbon dioxide by the DEPG does not represent the exact amount of captured gas, as there are losses in the solvent recovery column as well (Xin *et al.*, 2020). The issue of total power consumption in the capture section is also minimized by increasing the operating pressure of the absorption column. Despite requiring more energy to pressurize both the solvent and the incoming gas during the absorption process, the energy consumption is offset by the fact that less energy is needed in the stripping column, especially in the reboiler. Therefore, in all scenarios, DEPG regeneration accounts for the highest energy cost, being responsible for approximately 83% of the energy consumption in the CO₂ capture section, which is consistent with the findings of Belchior *et al.* (2020), who stated that solvent regeneration may account for more than 70% of the total electricity consumption in the CO₂ capture process. The unlisted elements pertain to the pumps connecting the two columns and complement the total power value.

Operating absorption at 120 bar, increasing the temperature from 25 °C to 37.77 °C significantly increased the energy consumed by the absorber components. The reflux from the bottom outlet was readjusted to 32.5%, whereas it was approximately 5% before, to minimize hydrogen loss. As a result, the tower not only operates at a higher temperature but also carries a greater mass of gas and solvent that needs to be heated. Hence, the solvent

quantity also increased considerably. However, in the end, the two energy consumptions become comparable, but a certain scenario is more attractive due to the lower solvent amount, considering that replenishing DEPG is an expensive process. Meanwhile, the lower pressure condition nearly doubled the total energy consumption and more than tripled the DEPG consumption. For this reason, it was determined that 120 bar and 25 °C would be the ideal scenario, even though it had a lower recovery compared to the others. The compensation in terms of energy and material savings is high, making it more financially advantageous.

Replicating the best condition found in the HB plant for the LAC plant yielded similar results in terms of the required amount of DEPG. In terms of capture and loss minimization, slightly greater success was achieved, as expected, and the energy cost was certainly lower. This can be attributed to the prior introduction of inert nitrogen in the conventional plants, while the LAC undergoes the entire pumping and heat exchange process without the presence of this component.

Table 4.3: CO₂ Capture Conditions Results.

Operational Conditions	CO₂ absorbed (%)	DEPG (kmol/h)	H₂ loss (kmol/h)	CO₂ recovered (%)
120 bar – 25 °C	0.9501	510.9	1.51	0.9383
120 bar – 37.77 °C	0.9494	839.0	0.19	0.9448
30 bar – 25 °C	0.9491	1,767.3	0.49	0.9422
LAC	0.9492	509	1.37	0.9405

Table 4.4: CO₂ Capture Energy Consumption.

Absorber Conditions	Energy Power (kW)		
	ABSORBER	STRIPPER	Total
120 bar – 25°C	18,775.6	65,636.4	115,963.2
120 bar – 37.77°C	24,199.0	49,904.0	107,933.6

30 bar – 25°C	9,422.3	112,161.5	211,998.9
LAC	9,518.3	58,892.3	99,921.54

4.5

Methanation

In HB plant, the methanation process was simulated under different conditions within literature references. It was observed that both carbon monoxide and carbon dioxide methanation reactions reached 100% conversion, consistent with the results reported by Gao et al. (2012). In the 35 bar scenario, CO methanation began to show a decline in conversion above 552 °C, which closely aligns with the 550 °C threshold identified in their study. This occurs due to the high equilibrium values until reaching higher temperatures, and due to the total absence of methane and slight residue of water, the former being consumed in its combustion and the latter separated at the bottom outlet of the stripping column condenser. Therefore, the chosen conditions were those with the lowest energy demand in the feed: 35 bar and 25 °C. The reactor provided the system with a useful power of 1020 kW due to the exothermic behavior. After a slight hydrogen loss following methanation, an adjustment was made to the amount of 390 kmol/h of air entering the second reformer. The value found to achieve balanced stoichiometric feed became 372.5 kmol/h (290.54 kmol/h of N₂) to produce 927.13 kmol/h of hydrogen, which decreased to 868.15 kmol/h after the methanation process.

4.6

Ammonia Synthesis

This section will present the optimization results of each stage of the two plants in the order of their respective simulation flowchart. In the common stages, each plant received an appropriate value to fulfill its purposes. Some stages are exclusive to each plant. All values, disturbances, and behaviors under the operational conditions are discussed.

4.6.1

Adiabatic single bed reactor

For the first case, it was possible to observe the growth of the conversion curve and temperature, as there is no heat exchange between the reactor bed and its surroundings (Figure 4.12). The growth in all cases begins exponentially, until the arrival of an inflection point, where the reaction starts to decelerate and converge to its maximum value. This inflection is associated with the decrease in the consumption and production rates of the components over the same length of the adiabatic reactor (Figure 4.13), justified by the brake caused by the increase in reactor temperature and the higher concentration of ammonia produced. Additionally, it is noticeable that the higher the pressure, the faster the equilibrium is reached, and the higher the final conversion. Thus, it is possible to reduce the reactor volume requirement, and consequently, the amount of catalyst. At 300 bar, the production of ammonia reaches its maximum point at around 4 m in length, achieving a conversion of 28.83%, while at 200 bar, it requires approximately twice the reactor volume of the previous case, reaching 26.17%. Finally, at 150 bar, a maximum of 24.36% is obtained, but it requires over 10 m in length, which was not sufficient to reach equilibrium. The conversion at 200 bar reached a value close to 20.6%, as cited in the study by Cloete et al. (2021), since reactor temperature and volume can lead to variations in the results. The best situation, or the one most consistent with thermodynamic equilibrium, is observed on the graph when the product formation rate reaches zero, which occurs near a length of 6 m for 300 bar (Figure 4.13).

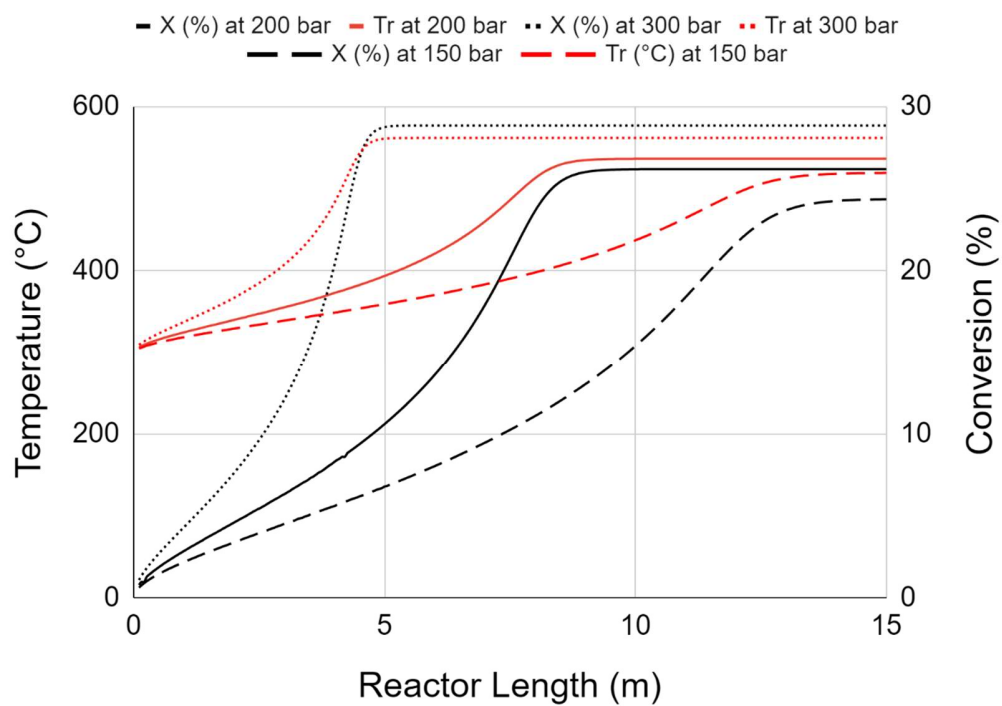


Figure 4.12: Temperature and Conversion behavior along reactor length of adiabatic fixed bed reactor (T_r = reactor temperature, X = conversion of ammonia).

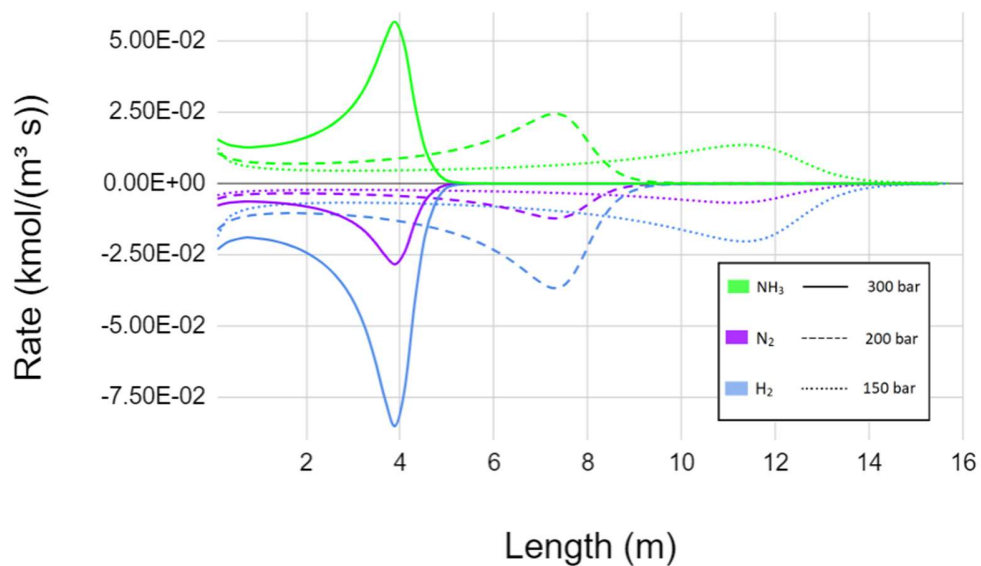


Figure 4.13: Rate of components along reactor length of adiabatic fixed bed reactor.

4.6.2

Adiabatic fixed bed reactor with looping

Given that a single adiabatic reactor bed alone is insufficient to achieve the significant and desired conversion levels, additional strategies must be implemented. The subsequent case studied utilizes the same adiabatic reactor bed but incorporates an ammonia looping system. Figure 4.14 illustrates the temperature behavior with looping under varying reactor operating pressures. Initially, the reactor outlet temperature increases as pressure rises, promoting higher conversion rates and consequently enhancing exothermic energy release. According to Figure 4.15, from pressures of approximately 170 bar onwards, it becomes possible to attain the theoretical maximum production of ammonia. Beyond this pressure threshold, what primarily changes are the recycle stream requirement and the necessary reactor volume. The simulation results indicate that as operating pressure increases, the required recycle stream decreases linearly, while the reactor volume necessary for achieving the targeted conversion decreases exponentially. Figure 4.16 provides insight into the substantial impact of operating at higher pressures; notably, at 300 bar, the system requires significantly less recycle and a considerably reduced reactor volume of only 23.69 m³ to attain the desired ammonia conversion.

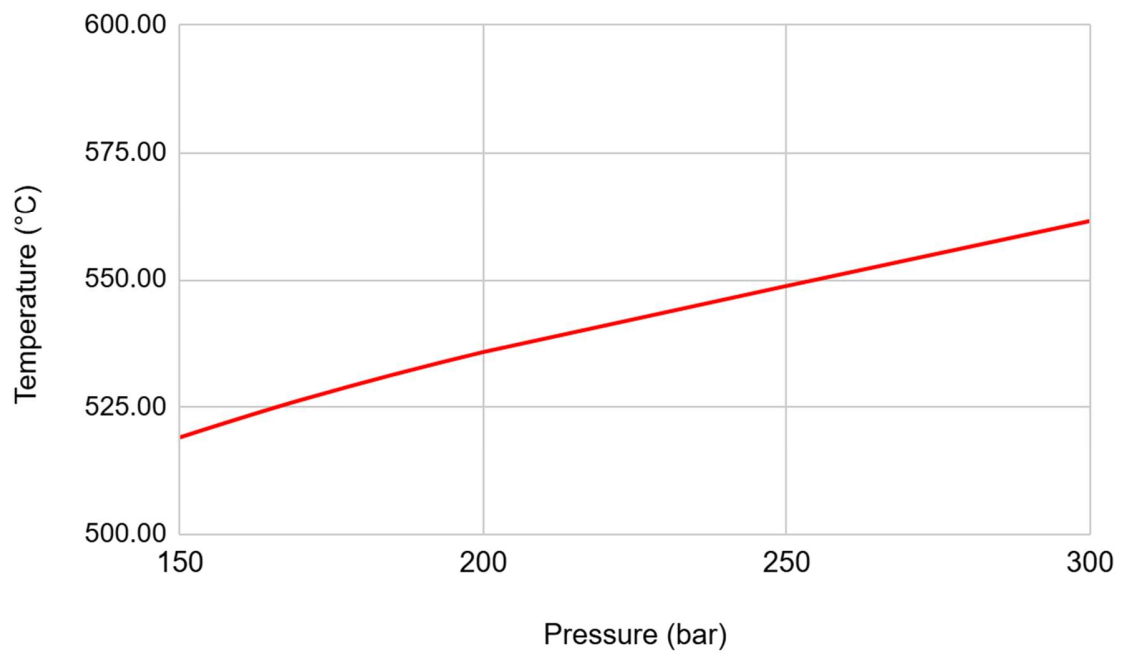


Figure 4.14: Temperature relation with pressure in adiabatic loop.

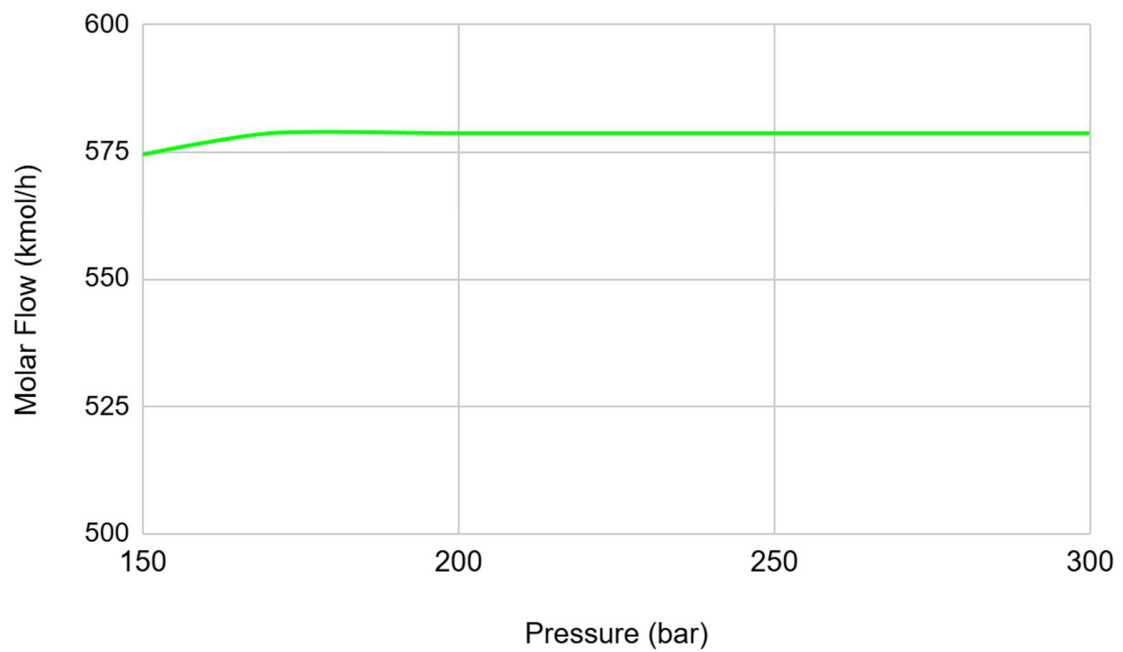


Figure 4.15: Ammonia molar flow relation with pressure in adiabatic loop.

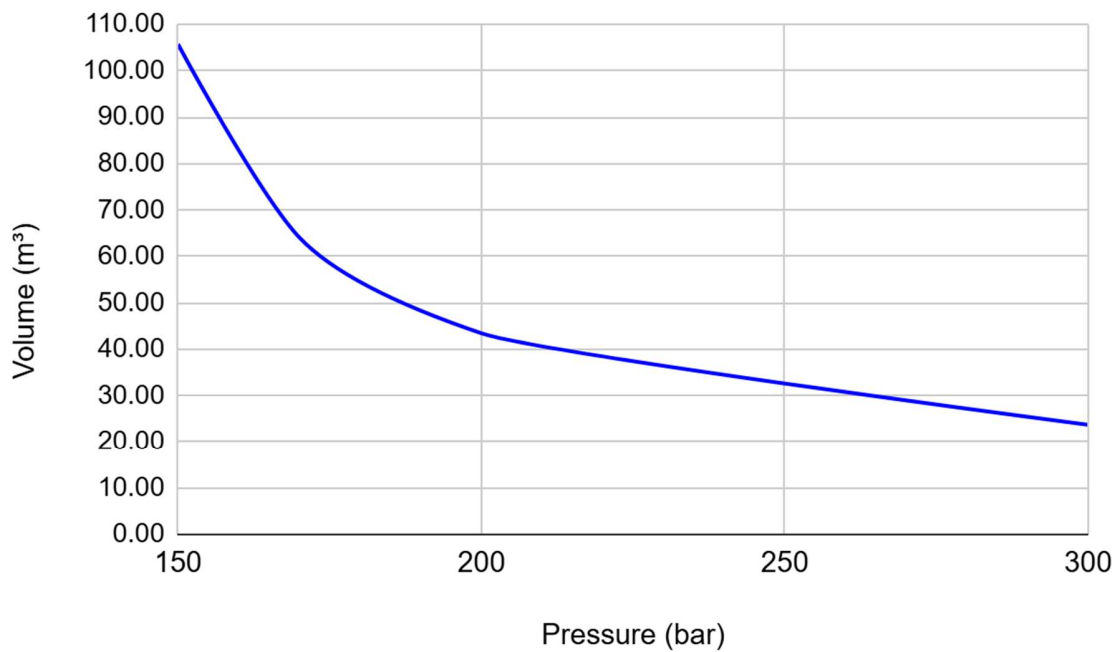


Figure 4.16: Reactor Volume relation with pressure in adiabatic loop.

4.6.3

Autothermal PFR fixed bed reactor

The results compare the values of a reactor with a volume equal to that of the first case with a 30 m³ reactor, which is the point at which the conversion begins to require a high volume to continue growing. The first graph shows the conversion behavior, and the first noteworthy aspect is that the higher inlet temperature curve is the uppermost, meaning that even though the reaction is exothermic, in this reactor, it is favored by kinetic factors for higher temperatures (Figure 4.17).

Another detail is that each volume set exhibits a similar behavior but differs in the outlet temperatures resulting in the highest conversion. For the maximum conversion values, as the inlet temperature increases, the necessary outlet temperature decreases, and it is noticeable that these spacings on the abscissa axis of the peaks gradually become larger. This occurs in both reactors; however, the larger one is more inclined to separate the curves as

mentioned. Another specific behavior of the 30 m³ reactor is that the T_{in} of 500 °C does not surpass all others throughout the graph. The second graph (Figure 4.18) shows the increase in heat released by the reactor when the feed has a higher temperature and exhibits a decreasing trend with the outlet temperature increase. However, in cases of lower T_{in} , the values fluctuate more and only significantly decrease after 500 °C for the outlet.

In summary, the graph presents the respective necessary outlet temperatures to achieve both the highest conversion and the highest released heat in kW (Figure 4.19). The best results found for the 6 m³ volume were with an inlet temperature of 500 °C and outlet temperature of 465 °C, achieving a conversion of 41.71%. As for the 30 m³ reactor, the best combination of inlet and outlet temperatures was 485 °C and 500 °C respectively, reaching a conversion of 53.51%. For both reactors, the highest energy released was from the scenario where the inlet temperature is 500 °C and the outlet temperature is 300 °C. The maximum conversion of the adiabatic case was 28.83%, used in the autothermal, requires only 3.13 m³ of reactor volume, which is a 47% reduction in size.

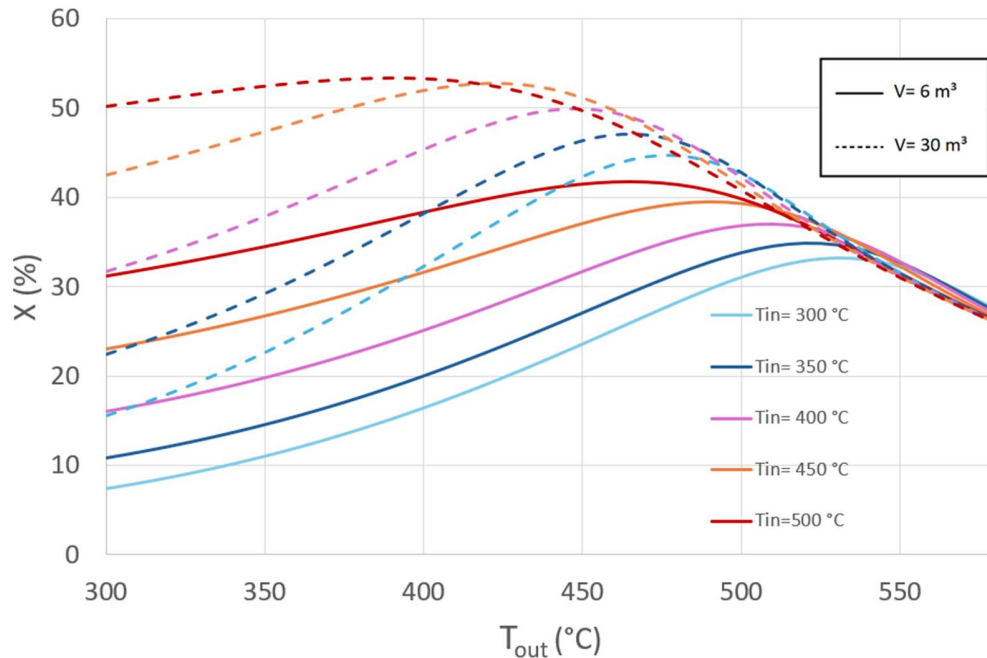


Figure 4.17: Conversion with different inlet and outlet temperatures in autothermal reactor.

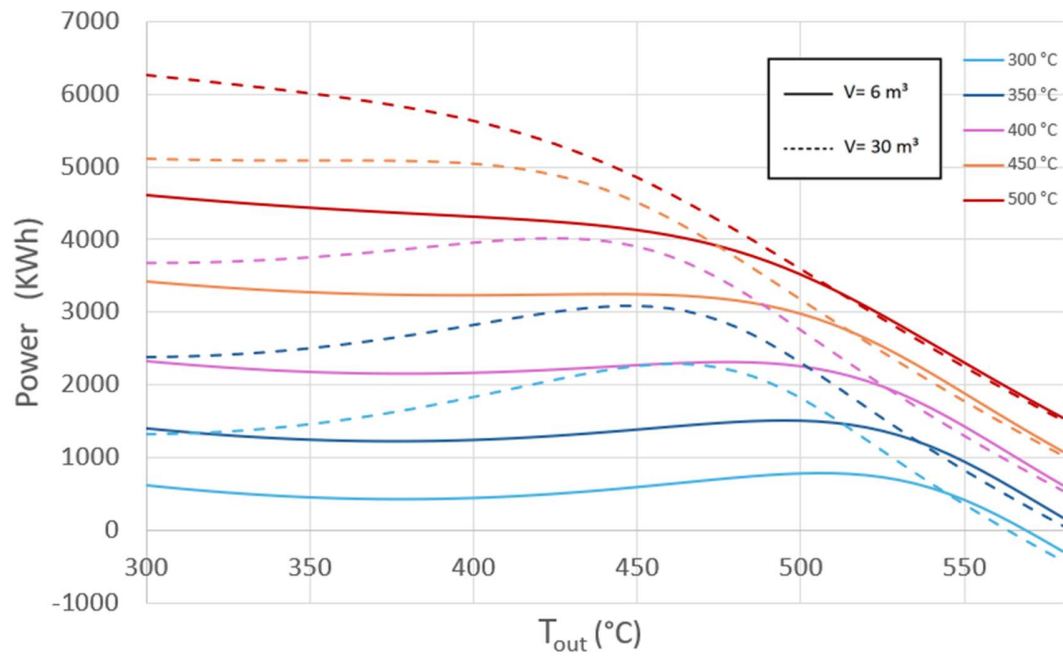


Figure 4.18: Power with different inlet and outlet temperatures in autothermal reactor.

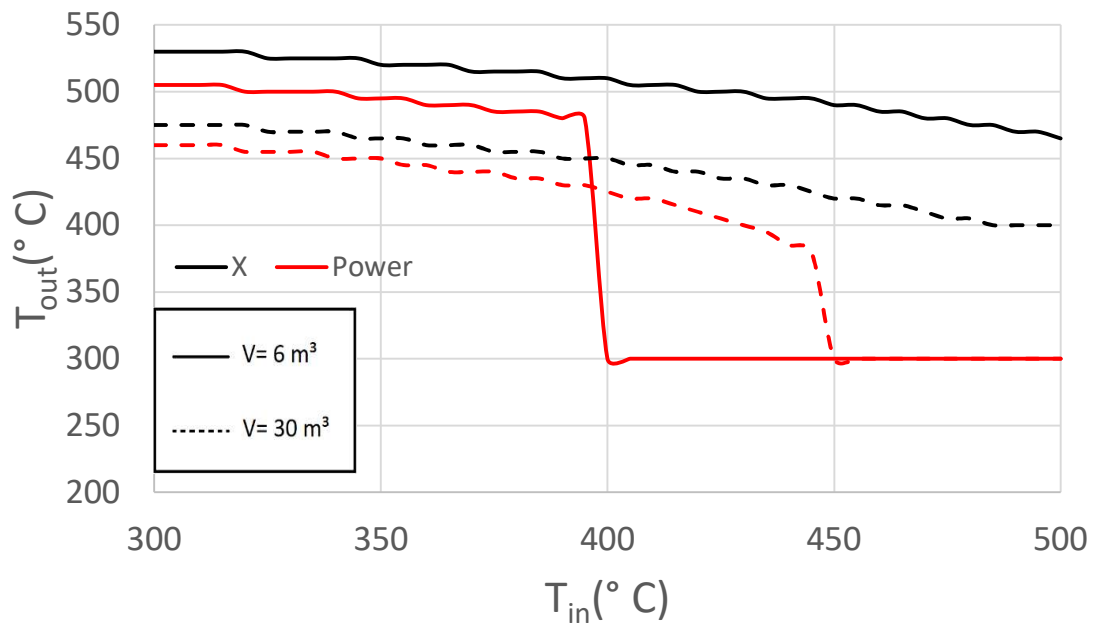


Figure 4.19: Combination of inlet and outlet temperatures to achieve greater conversion and power produced.

4.6.4

Autothermal fixed bed reactor with looping

The maximum conversion in a 30 m³ isobaric reactor is 53.51%. To achieve the same conversion, it was necessary to add looping and adjust the reactor volume, requiring 56.39 m³. It is evident that, in this case, the volume demand increases significantly compared to that of adiabatic looping, as the reflux does not adjust sufficiently to meet the reactor volume requirement (Figure 4.20). Therefore, for pressures below 180 bar, the resulting volumes reached extrapolated values that could not be represented in the graphic.

For this case, to have a conversion reactor of 28.83%, equals to the adiabatic reactor loop, keeping the reflux constant, 10.88 m³ were enough and it represents a 40.7% volume reduction.

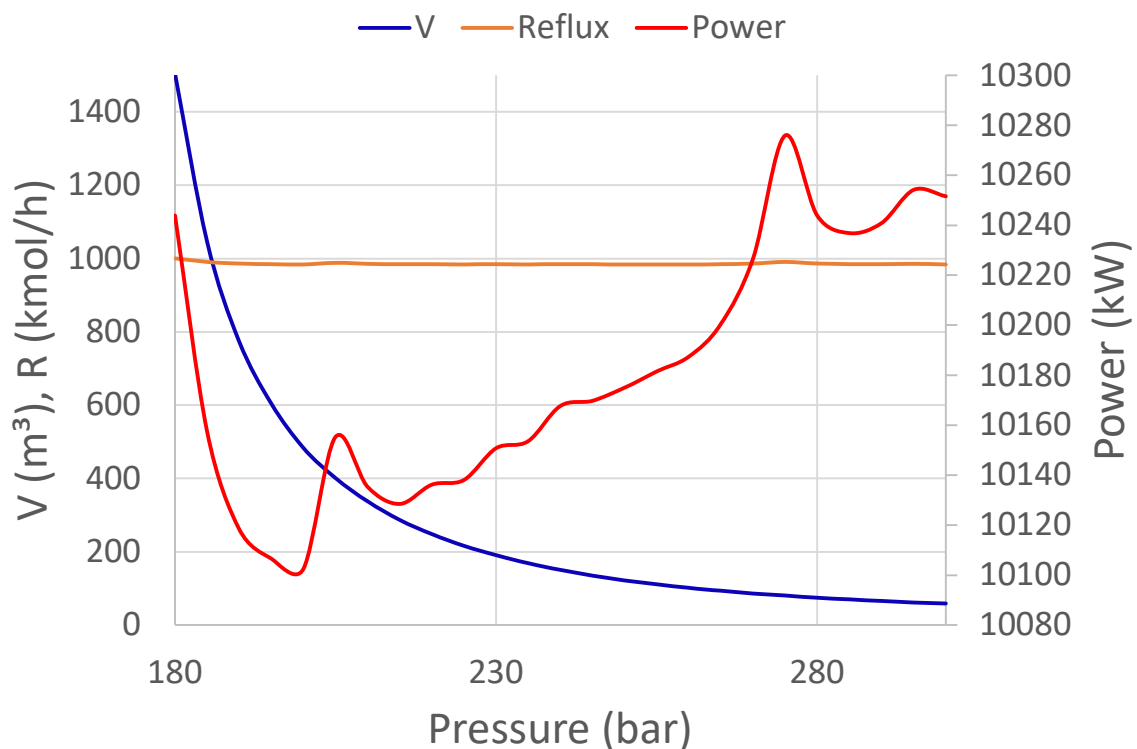


Figure 4.20: Volume, Reflux and Power relation with pressure in autothermal loop.

4.6.5

Adiabatic with intercooling system

The results of arranging multiple fixed beds in series, alternating them with a cooling system were observed. Another strategy involved simulating the reaction in a reactor with a heat exchanger inserted inside. By manipulating the outlet temperature, it was possible to determine the necessary volumes for each catalyst (Figure 4.21). In this figure, it can be observed that the overall conversion increases rapidly within 2.8 m^3 of catalyst in the first reactor bed to achieve conversions exceeding 28%. Upon passing through the initial cooling process and the second bed, the graph indicates that an additional 34.1 m^3 of catalyst is required to reach the overall conversion value of 43.06%. Therefore, a notable increase in the catalyst volume requirement and a significant reduction in the overall conversion rate are evident as one progresses through another sequence of a cooler followed by a reactor bed.

The third bed of the reactor, for example, has a very high volume, and the fourth one even larger. Therefore, achieving high conversions in this process above 50% already makes this strategy quite expensive in terms of investment. Ultimately, a global conversion of 60% is achieved, which is not ideal. In an attempt to improve this strategy and reduce the demand for reactor volume, we sought to work with excess of nitrogen. The results obtained were satisfactory, as the final conversion in the last case was achieved already in the second bed (Figure 4.22). Additionally, an ammonia looping was planned to convert any remaining reactants; however, the simulator revealed limitations when processing the results with an excess stream. To accurately represent this scenario, two reactor beds were implemented, followed by a flash separator and subsequently another pair of reactors. This new configuration allowed achieving a conversion of 90%.

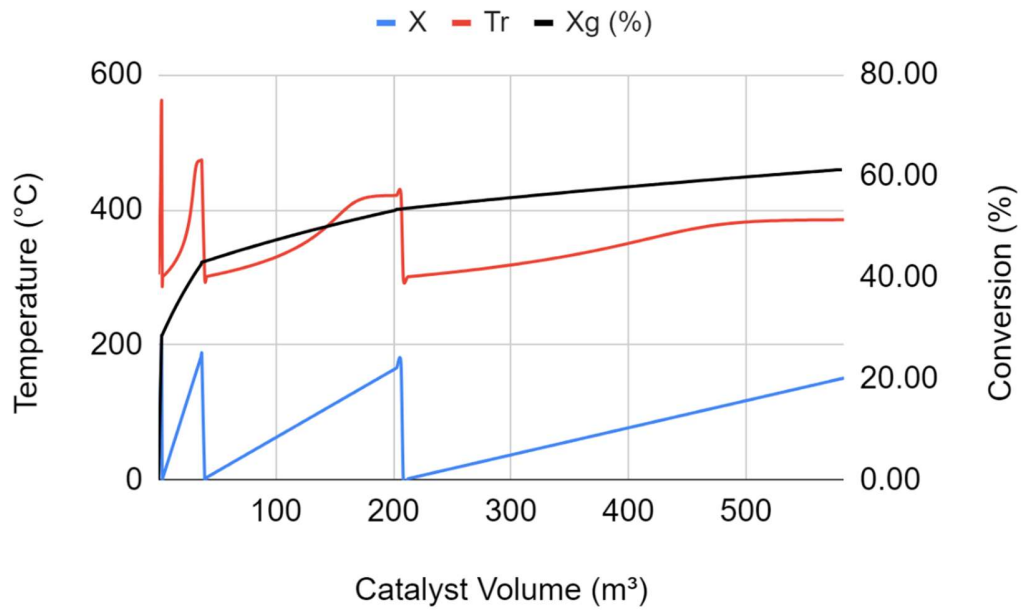


Figure 4.21: Temperature and Conversion in adiabatic reactor series with intercooling along catalyst bed.

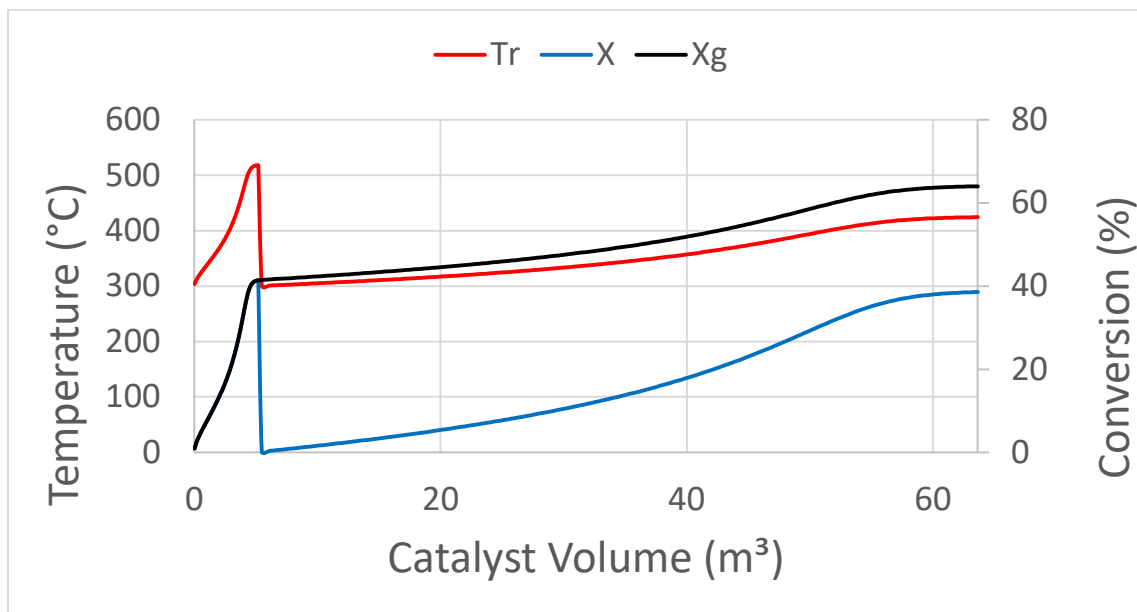


Figure 4.22: Temperature and Conversion in adiabatic reactor series with intercooling along catalyst bed with nitrogen excess.

4.6.6

Excess of Nitrogen

The optimal nitrogen excess was determined to be approximately 120% relative to the hydrogen excess. To systematically increase the excess nitrogen as per the established methodology, adjustments to the reactor volume were necessary in order to maintain constant residence time. However, the residence time is a function of the molar flow rate that decreases along the reactor, which makes its utilization complex. In order to simplify, spatial time was employed as an approximation. At 360% of nitrogen excess, the conversion continues to reach its maximum and then starts to slightly decrease, whereas at 120% of nitrogen excess, a value close to the maximum can already be found without needing to increase the volume by much (Figure 4.23).

With the presence of three reactors in series separated by flash tanks that separate the product from the unreacted, the reactors were respectively modeled at 35.63 m³, 17.88 m³, and 13.38 m³ using the simulator's optimizer tool. The nitrogen input quantity for this series of reactors was then reevaluated, resulting in 392% excess nitrogen for the second reactor and 908% for the third, as the excess enters the first reactor. Therefore, the solution was to reassess the quantity entering from the first reactor. From the performance graph, it can be concluded that with 100 kmol/h of excess, the overall conversion reaches its maximum value of 94.33% (Figure 4.24). It is important to note that this series of reactors is similar to a looping strategy; however, due to issues with Hysys convergence caused by disproportionate hydrogen excess within the looping, this strategy was used to approach the case.

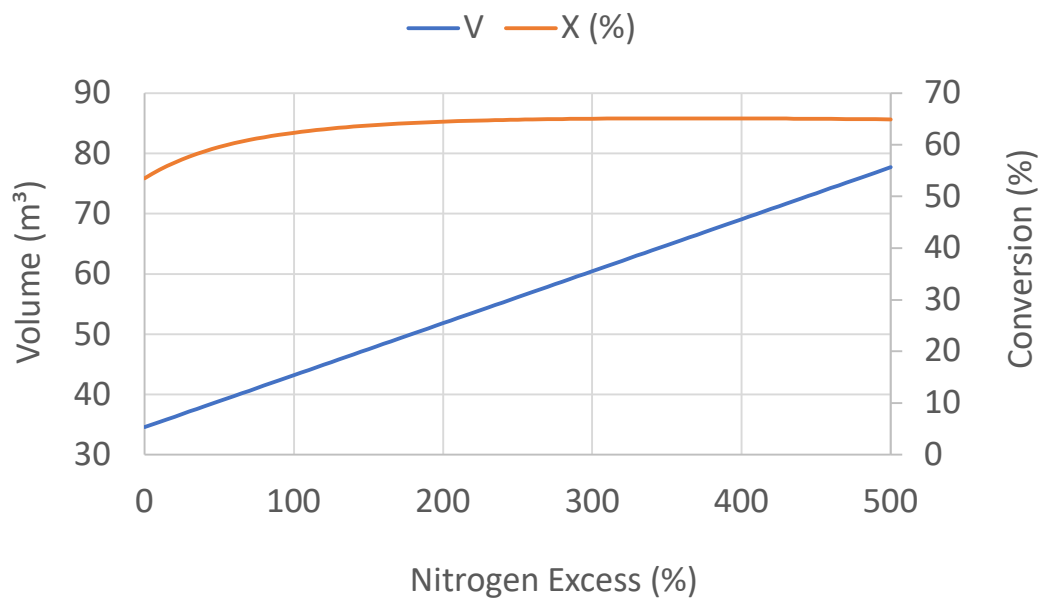


Figure 4.23: Effect of Nitrogen Excess in single autothermal bed.

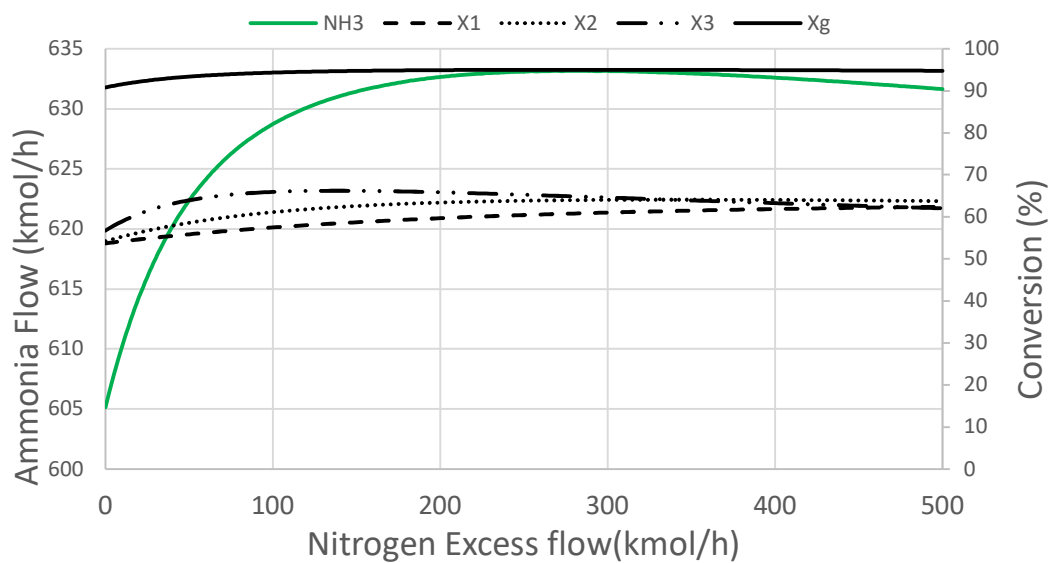


Figure 4.24: Effect of Nitrogen Excess in autothermal reactor series.

4.7

Utilities and Energy consumption

For the HB plant, Tables 4.5 and 4.6 show, respectively, the amounts of each type of hot and cold utility streams, expressed in kJ/h. Furthermore, these tables detail the potential energy savings ("Saving Potential") and the actual utility consumption after energy integration ("Target") and the "Energy Cost" which represents how much the "Current" energy exceeds the "Saving Potential". The same procedure has been applied to the LAC plant, with corresponding results presented in Tables 4.7 and 4.8. The energy consumption associated with each process stage, including a detailed breakdown of the contribution from each utility stream, is depicted in Figures 4.25 and 4.26 by means of a Sankey diagram.

Table 4.5: Hot utilities, energy saving and target energy from HB.

Utility type	Current [kJ/h]	Target [kJ/h]	Saving Potential [kJ/h]	Energy Cost Savings [%]	$\Delta T_{\min}$ [° C]
LP Steam	2.63E+05	0	2.63E+05	100	10
Fired Heat (1000)	1.85E+08	1.04E+08	8.12E+07	43.82	25
Hot Oil	2.48E+06	0	2.48E+06	100	5
Very High Temperature	5.92E+07	0	5.92E+07	100	10
Total Hot Utilities	2.47E+08	1.04E+08	1.43E+08	66.56	-

Table 4.6: Cold utilities, energy saving and target energy from HB.

Utility type	Current [kJ/h]	Target [kJ/h]	Saving Potential [kJ/h]	Energy Cost Savings [%]	$\Delta T_{\min}$ [° C]
Refrigerant 1	1.13E+08	1.30E+07	1.00E+08	88.47	3
Refrigerant 3	4.91E+07	2.06E+06	4.71E+07	95.81	2
Cooling Water	1.10E+08	8.45E+06	1.02E+08	92.32	5
LP Steam Generation	1.24E+07	4.94E+07	-3.69E+07	296.95	10
MP Steam Generation	6.60E+07	5.58E+07	1.02E+07	-15.44	10
Very Low Temperature	4.17E+07	1.00E+06	4.07E+07	97.59	2
Air	0	1.05E+08	-1.05E+08	-	10
HP Steam Generation	0	1.05E+07	-1.05E+07	-	10
Refrigerant 2	0	6.71E+05	-6.71E+05	-	3
Refrigerant 4	0	3.07E+06	-3.07E+06	-	2
Total Cold Utilities	3.92E+08	2.49E+08	1.43E+08	118.75	-

Table 4.7: Hot utilities, energy saving and target energy from LAC.

Utility type	Current [kJ/h]	Target [kJ/h]	Saving Potential [kJ/h]	Energy Cost Savings [%]	$\Delta T_{\min}$ [° C]
Fired Heat (1000)	1.65E+08	8.58E+07	7.94E+07	48.06	25
Refrigerant 1 Generation	5.44E+05	0	5.44E+05	-100	3
Fired Heat (2000)	1.98E+07	0	1.98E+07	100	30
Very High Temperature	8.59E+07	0	8.59E+07	100	10

Total Hot Utilities	2.71E+08	8.58E+07	1.86E+08	77.07	-
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Table 4.8: Cold utilities, energy saving and target energy from LAC.

Utility type	Current [kJ/h]	Target [kJ/h]	Saving Potential [kJ/h]	Energy Cost Savings [%]	$\Delta T_{\min}$ [° C]
Refrigerant 1	1.67E+08	1.21E+07	1.55E+08	92.75	3
Refrigerant 3	8.25E+07	8.01E+06	7.45E+07	90.29	5
Cooling Water	1.08E+07	2.91E+07	-1.83E+07	169.62	10
LP Steam Generation	8.70E+07	4.84E+07	3.86E+07	-44.4	10
MP Steam Generation	6.20E+06	2.08E+06	4.13E+06	66.53	2
Very Low Temperature	4.98E+07	1.13E+06	4.87E+07	97.73	2
Air	0	1.01E+08	-1.01E+08	-	10
HP Steam Generation	0	1.35E+07	-1.35E+07	-	10
Refrigerant 2	0	9.09E+05	-9.09E+05	-	3
Refrigerant 4	0	1.39E+06	-1.39E+06	-	2
Total Cold Utilities	4.03E+08	2.18E+08	1.86E+08	115.85	-

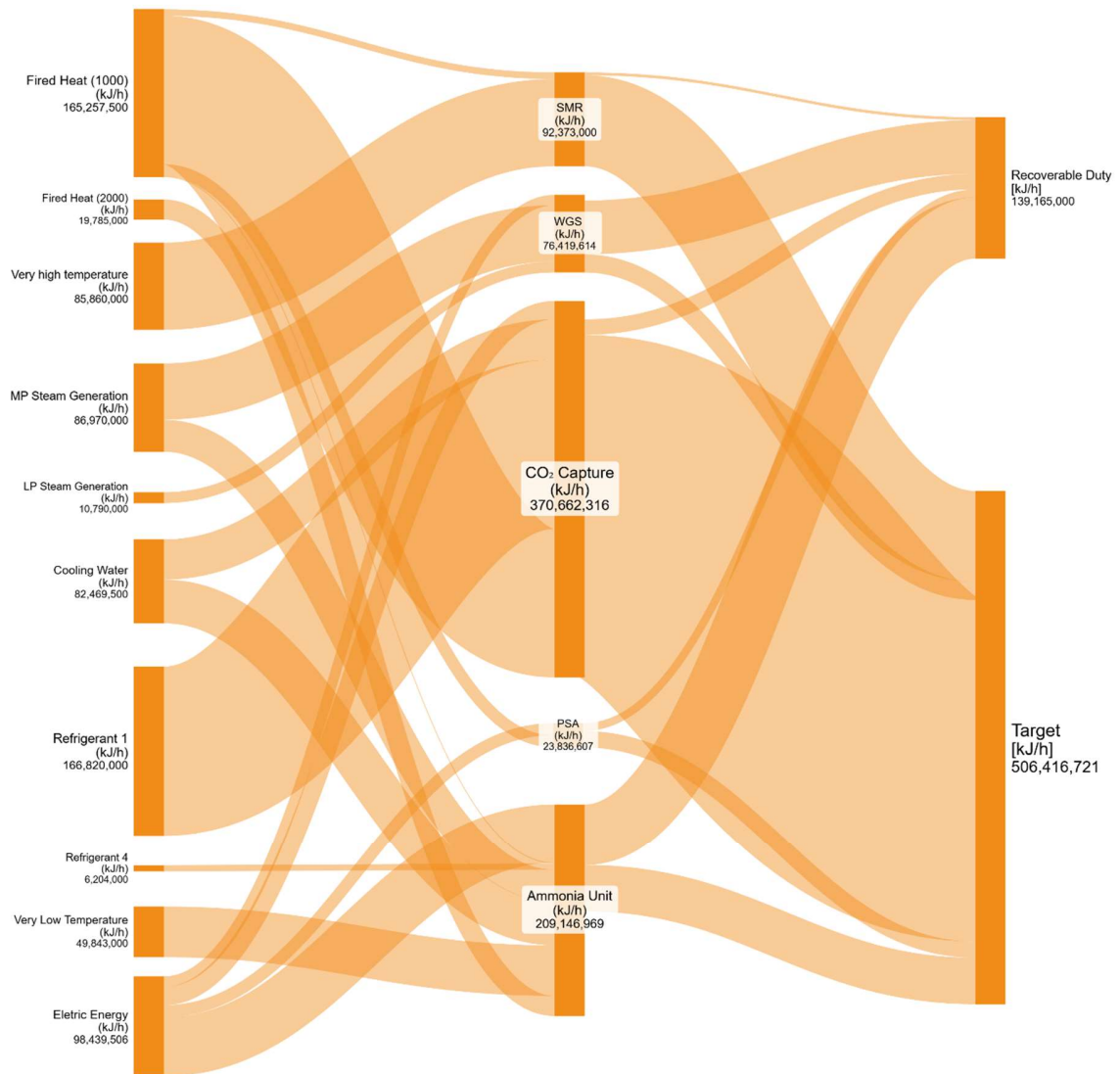


Figure 4.25: Sankey Diagram of energy distribution in HB.

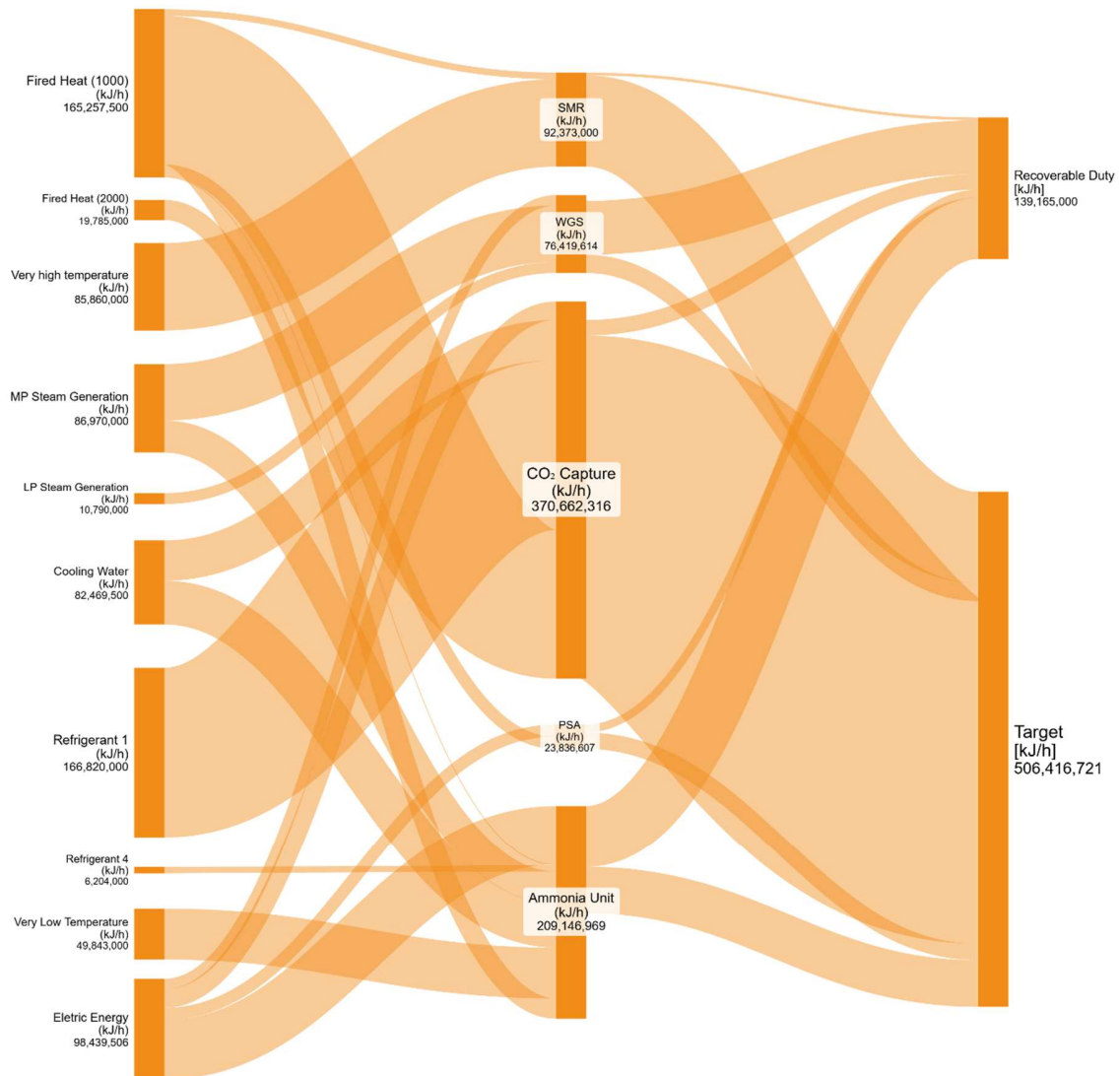


Figure 4.26: Sankey Diagram of energy distribution in LAC.

Based on the results obtained from the energy integration analysis, Aspen HYSYS generates graphical representations that compare energy consumption without (Actual) and with (Target) electrical energy integration for both hot and cold utilities within the plants. These graphical results are illustrated in Figure 4.25 for the HB plant and in Figure 4.26 for the LAC plant. By combining the utilities' energy demand with electrical energy consumption, the total energy expenditure for each entire process is summarized in Tables 4.9 and 4.10. The LAC plant incurred the highest overall energy expenditure, totaling 4.56×10^8 kJ/h; despite achieving higher energy reuse in utility streams, this plant exhibited

significantly higher electrical energy consumption, amounting to 1.48×10^8 kJ/h. Conversely, the HB plant presented lower levels of energy reuse but benefited from lower electrical energy consumption of 5.52×10^7 kJ/h, resulting in a total energy expenditure of 4.08×10^8 kJ/h.

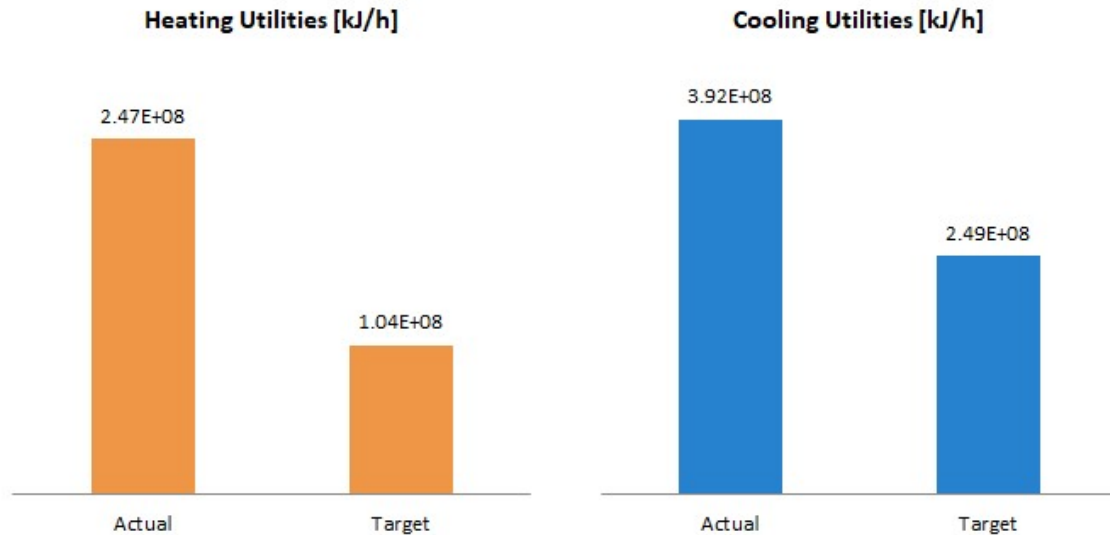


Figure 4.27: HB utilities consumption.

Table 4.9: HB - Utilities and energy consumption.

Property	Actual	Target	Saving Energy (%)
Heating Utilities (kJ/h)	2.47E+08	1.04E+08	57.88
Cooling Utilities (kJ/h)	3.92E+08	2.49E+08	36.5
Electric Power (kJ/h)	5.52E+07	-	-
Total Power (kJ/h)	6.94E+08	4.08E+08	-

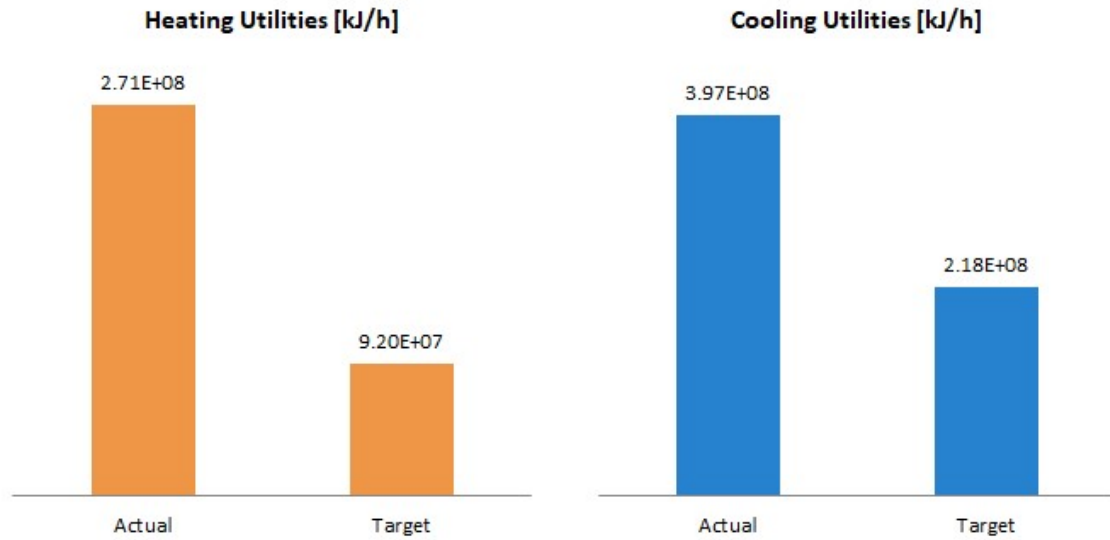


Figure 4.28: LAC utilities consumption.

Table 4.10: LAC - Utilities and energy consumption.

Property	Actual	Target	Saving Energy (%)
Heating Utilities (kJ/h)	2.71E+08	9.20E+07	66.09
Cooling Utilities (kJ/h)	3.97E+08	2.18E+08	45.20
Electric Power (kJ/h)	1.47E+08	-	-
Total Power (kJ/h)	8.15E+08	4.56E+08	-

4.8

Overall comparison between HB and LAC

This chapter provides a comprehensive comparative analysis of the results obtained from simulating and optimizing the conventional Haber-Bosch (HB) process and the alternative Linde Ammonia Concept (LAC) process for ammonia synthesis. The primary distinctions between these two methodologies begin at the reforming stage. The HB process

employs a two-step reforming approach, utilizing a steam methane reformer (SMR) operating at 800°C and 20 bar, followed by an autothermal reformer optimized at 950°C and 30 bar. Conversely, the LAC process achieves higher methane conversion (98.76%) through a single-step high-temperature SMR operated at 1000°C and 20 bar, significantly higher than the HB's conversion rate of 76.73%. Despite this advantage, the high-temperature operation of the LAC reformer translates into greater initial energy consumption and higher overall utility demand.

Further differences emerged in the Water Gas Shift (WGS) reaction stage. The HB plant operated with higher inlet temperatures in both its High-Temperature Shift (HTS, 270°C) and Low-Temperature Shift (LTS, 170°C) reactors, resulting in outlet temperatures of approximately 352.16°C and 181.38°C, respectively. The LAC plant, benefiting from a streamlined process configuration, employed lower inlet temperatures for HTS (240°C) and LTS (162.5°C), yet achieved comparable outlet temperatures of 351.68°C and 182.45°C. This slightly lower operational temperature requirement contributed to enhanced energy management within the LAC plant.

In the CO₂ capture stage, both plants employed DEPG as the solvent in absorption columns operating under similar conditions of 120 bar and 25°C. However, the LAC plant demonstrated marginally better efficiency, consuming slightly less solvent (509 kmol/h compared to 510.9 kmol/h for HB) and achieving reduced hydrogen losses (1.37 kmol/h versus 1.51 kmol/h in HB), thus offering modest operational advantages.

Significant operational differences were particularly notable in the ammonia synthesis reactor configurations. The HB plant used an autothermal reactor integrated with ammonia looping, reaching a maximum ammonia conversion of around 53.51% with a reactor volume of 56.39 m³. Conversely, the LAC process incorporated a sophisticated configuration comprising three reactors arranged in series, each separated by flash separation and utilizing a nitrogen-excess strategy. This advanced setup allowed the LAC process to achieve a significantly higher ammonia conversion rate of 94.33%, though it required a larger total reactor volume of 66.88 m³. The considerable increase in conversion efficiency

presented by the LAC process clearly compensates for the slightly larger reactor and catalyst volume requirement.

A thorough analysis of energy consumption also highlighted critical differences. Initially, the LAC plant required around 4% more total utility energy compared to the HB process (2.71×10^8 kJ/h versus 2.47×10^8 kJ/h), corresponding to about a 4% difference. Nonetheless, despite its higher initial utility demand, the LAC plant showed significantly superior thermal energy integration, reducing hot utility requirements by 66.09%, compared to the HB's 57.88%, thus highlighting its superior energy integration efficiency. Likewise, the LAC plant's cold utility consumption decreased by approximately 45.20%, significantly better than the 36.5% reduction observed in the HB plant. Despite these efficiency gains, the LAC plant exhibited substantially higher electricity usage, approximately 2.6 times greater than the HB plant (1.47×10^8 kJ/h versus 5.52×10^7 kJ/h), primarily due to the operation of Pressure Swing Adsorption (PSA) units and an Air Separation Unit (ASU), underscoring its significant electrical energy demand.

Combining all energy aspects, including thermal and electrical demands, revealed that the HB plant had lower overall energy consumption (4.08×10^8 kJ/h) compared to the LAC plant (4.56×10^8 kJ/h). However, this higher overall consumption in the LAC plant was offset by its significantly greater ammonia productivity, producing 665.1 kmol/h compared to the HB plant's 578.0 kmol/h. On a specific basis, the LAC plant also exhibited a slightly lower energy consumption of 40.25 GJ per tonne of ammonia produced, in contrast to 41.46 GJ/t observed in the HB process, further highlighting the efficiency advantage of the LAC configuration.

From an overall viability perspective, the choice between the HB and LAC processes depends on prioritizing either reduced energy cost or maximized production capacity. While the HB approach provides advantages in terms of lower energy consumption, especially electricity, the higher ammonia production and superior energy integration efficiency make the LAC process particularly appealing. This appeal can be further enhanced if integrated with renewable energy sources, thereby addressing its primary disadvantage related to higher electricity consumption. Consequently, the LAC approach emerges as a more contemporary

and potentially more viable alternative for ammonia production when factoring in comprehensive energy, economic, and environmental considerations.

5. Conclusion

This dissertation has presented a comprehensive comparative study between the conventional Haber-Bosch (HB) route and the alternative Linde Ammonia Concept (LAC) for ammonia production. Through rigorous modeling and simulation in Aspen HYSYS, each stage of the processes—reforming, shift conversion, CO₂ capture, hydrogen purification, and ammonia synthesis—was evaluated in detail under kinetic and thermodynamic frameworks. These simulations allowed for the development of sensitivity analyses and optimization studies that provided a robust understanding of process behavior under a range of operating conditions.

The initial objectives were systematically addressed. Main reactions in each unit operation were modeled using relevant thermodynamic and kinetic parameters. Sensitivity surfaces were constructed to map the effects of key operational variables on conversion, yield, and energy demand. These insights facilitated the optimization of process conditions and reactor configurations. Particularly, ammonia synthesis was investigated through several setups—adiabatic single-bed, autothermal, multi-bed with intercooling, and configurations employing nitrogen excess. The LAC process, which leverages a Pressure Swing Adsorption (PSA) unit for hydrogen purification and an Air Separation Unit (ASU) for nitrogen supply, enabled the successful implementation of the nitrogen excess strategy. This resulted in a reduction in reactor volume requirements without compromising conversion or throughput.

The analysis of CO₂ absorption under various conditions confirmed the energetic and operational advantages of the LAC configuration. Furthermore, while the Haber-Bosch route relies on energy-intensive methanation for gas purification, the LAC configuration benefits from a simplified flow scheme with PSA technology, reducing both the number of unit operations and the associated energy requirements.

From a performance standpoint, the LAC process proved superior. It delivered a higher ammonia production rate of 665.1 kmol/h compared to 578.0 kmol/h for the HB route, while also exhibiting a lower specific energy consumption of 40.25 GJ/t NH₃ versus 41.46 GJ/t NH₃. These findings clearly demonstrate that the LAC route is not only more energy-

efficient but also more productive under the studied conditions. The integration of modern separation and purification technologies, combined with reactor design strategies such as looping and nitrogen excess, contribute significantly to this enhanced performance.

It is important to note that the energy integration performed in this study was limited to thermal energy across heat exchangers and reaction stages. Therefore, future work should focus on extending this integration to include mechanical energy streams—specifically, the coupling and optimization of compressors, pumps, and expanders. Such an approach could yield further improvements in process efficiency and contribute to a more holistic energy optimization framework.

Additionally, while this study emphasized energy consumption and productivity, it did not include a formal exergy analysis. Future research should incorporate a detailed exergy analysis of both HB and LAC routes to quantify irreversibilities and better assess the thermodynamic efficiency of each process. This would provide deeper insight into the quality of energy used and the potential for further process intensification. Combined with economic and environmental assessments, such studies could guide the development of even more sustainable ammonia production technologies in the future.

6.

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7. Appendices

Energy integration of HB plant									
Heat Exchanger	Type	Base Duty [kJ/h]	Recoverable Duty [kJ/h]	Hot T _{in} [°C]	Hot T _{out} [°C]	Cold T _{in} [°C]	Cold T _{out} [°C]	Hot Side Fluid	Cold Side Fluid
Condenser	Cooler	7.22E+07	0.0	141.6	3.2	-25	-24	To Condenser@COL2_TO_Reflux@COL2	Refrigerant 1
E-101	Heater	6.51E+06	2.71E+06	1000	400	37.8	520	Fired Heat (1000)	2_To_3
E-102	Cooler	8.33E+06	8.33E+06	929.8	795	174	175	6_To_7	MP Steam Generation
E-103	Cooler	9.24E+05	9.24E+05	626.9	550	174	175	9_To_10	MP Steam Generation

E-104	Cooler	4.81E+07	4.77E+07	950	270	174	175	13_To_14	MP Steam Generation
E-105	Cooler	1.24E+07	5.25E+06	353.2	170	124	125	15_To_16	LP Steam Generation
E-106	Cooler	4.91E+07	1.10E+07	428.9	25	-65	-64	18_To_19	Refrigerant 3
E-107	Cooler	6.73E+05	0.0	218.6	25	20	25	24_To_26	Cooling Water
E-108	Heater	2.63E+05	2.63E+05	125	124	24.8	25	LP Steam	27_To_28@TP L1
E-109	Cooler	1.09E+08	8.37E+06	293.6	25	20	25	32_To_33	Cooling Water
E-110	Heater	2.48E+06	3.39E+05	280	250	-43.6	25	Hot Oil	41_To_42
E-111	Heater	1.73E+07	2.97E+06	1000	400	220.6	485	Fired Heat (1000)	47_To_48
E-112	Cooler	4.17E+07	7.46E+06	400	-109.4	-270	-269	49_To_50	Very Low Temperature
Q-103	Heater	5.92E+07	0.0	3000	2999	520	800	Very High Temperature	Q-103@Main
Q-108	Cooler	8.61E+06	8.61E+06	950	949.5	174	175	Q-108	MP Steam Generation
Q-125	Cooler	3.71E+07	3.71E+07	485	400	-25	-24	Q-125	Refrigerant 1
Q-126	Cooler	3.67E+06	0.0	25	24.5	-25	-24	Q-126	Refrigerant 1

Reboiler	Heater	1.62E+08	0.0	1000	400	265.9	283.1	Fired Heat (1000)	To Reboiler@COL 2_TO_Boilup@ COL2
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Energy integration of LAC plant									
Heat Exchanger	Type	Base Duty [kJ/h]	Recoverable Duty [kJ/h]	Hot T _{in} [°C]	Hot T _{out} [°C]	Cold T _{in} [°C]	Cold T _{out} [°C]	Hot Side Fluid	Cold Side Fluid
Condenser	Cooler	6.19E+07	0.0	114.1	-17.5	-25	-24	To Condenser@COL2_TO_Reflux@COL2	Refrigerant 1
E-101	Heater	6.51E+06	2.77E+06	1000	400	37.8	520	Fired Heat (1000)	2_To_3
E-102	Cooler	5.54E+07	5.31E+07	1159.4	240	174	175	6_To_7	MP Steam Generation
E-103	Cooler	1.08E+07	4.06E+06	351.7	162.5	124	125	8_To_9'	LP Steam Generation
E-104	Cooler	3.88E+07	8.88E+06	427.9	25	20	25	11_To_12	Cooling Water
E-105	Cooler	6.96E+05	0.0	222.8	25	20	25	16_To_17'	Cooling Water
E-106	Cooler	1.05E+08	6.44E+06	294.8	25	-25	-24	23_To_24	Refrigerant 1
E-107	Heater	1.19E+07	7.80E+06	1000	400	25.3	400	Fired Heat (1000)	31'_To_32

E-108	Cooler	6.20E+06	5.25E+06	447.7	250	-103	-102	36_To_37	Refrigerant 4
E-109	Heater	6.25E+05	6.25E+05	1000	400	250	270	Fired Heat (1000)	37_To_38'
E-110	Cooler	3.16E+07	3.16E+07	1144.1	485	174	175	45_To_46	MP Steam Generation
E-111	Cooler	2.94E+07	5.16E+06	400	-109.5	-270	-269	47_To_48	Very Low Temperature
E-112	Heater	1.33E+07	8.61E+06	2000	400	-109.5	485	Fired Heat (2000)	49_To_50
E-113	Cooler	1.38E+07	2.44E+06	400	-109.5	-270	-269	51_To_52	Very Low Temperature
E-114	Heater	6.52E+06	4.25E+06	2000	400	-109.5	485	Fired Heat (2000)	53_To_54
E-115	Cooler	6.66E+06	1.19E+06	400	-109.5	-270	-269	55_To_56	Very Low Temperature
E-116	Heater	5.44E+05	5.44E+05	-24	-25	-109.5	-40	Refrigerant 1 Generation	57_To_58
Q-103	Heater	8.59E+07	0.0	3000	2999	520	1000	Very High Temperature	Q-103
Q-125	Cooler	2.59E+07	2.59E+07	485	400	20	25	Q-125	Cooling Water

Q-128	Cooler	1.19E+07	1.19E+07	485	400	20	25	Q-128	Cooling Water
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